



Project-2- Deployment of Online Ticketing and Event Management Applications and Infrastructure

Domain: E-Commerce and Entertainment Project Overview: BookMyShow is a popular online movie ticketing platform that allows users to browse movies, book tickets, and manage their bookings. The company is transitioning from a monolithic architecture to a microservices-based architecture to improve scalability, deployment efficiency, and infrastructure management. To achieve this, BookMyShow has decided to adopt DevOps practices with CI/CD pipelines, containerization, and automated infrastructure provisioning. The project will use AWS as the primary cloud provider for hosting servers, databases (if any), and deploying applications.

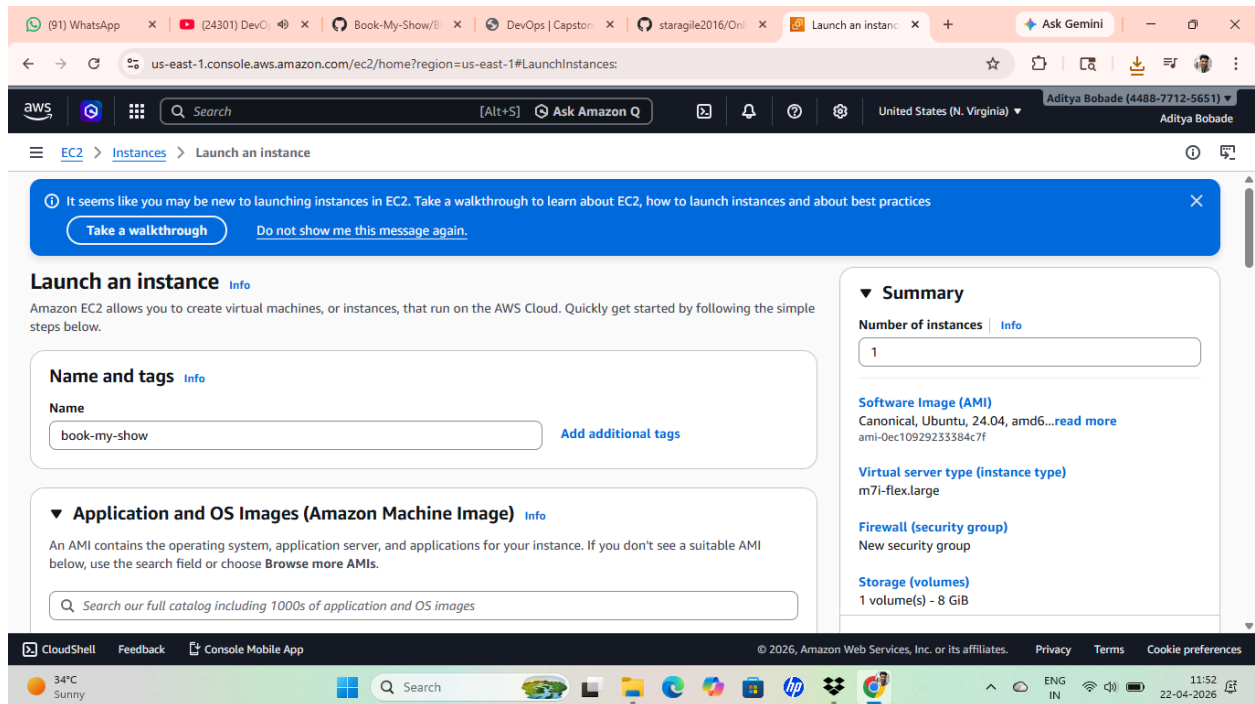
GitHub URL: <https://github.com/adityabob/OnlineTicketPlatform-BMS>

Batch Code: SA2512007

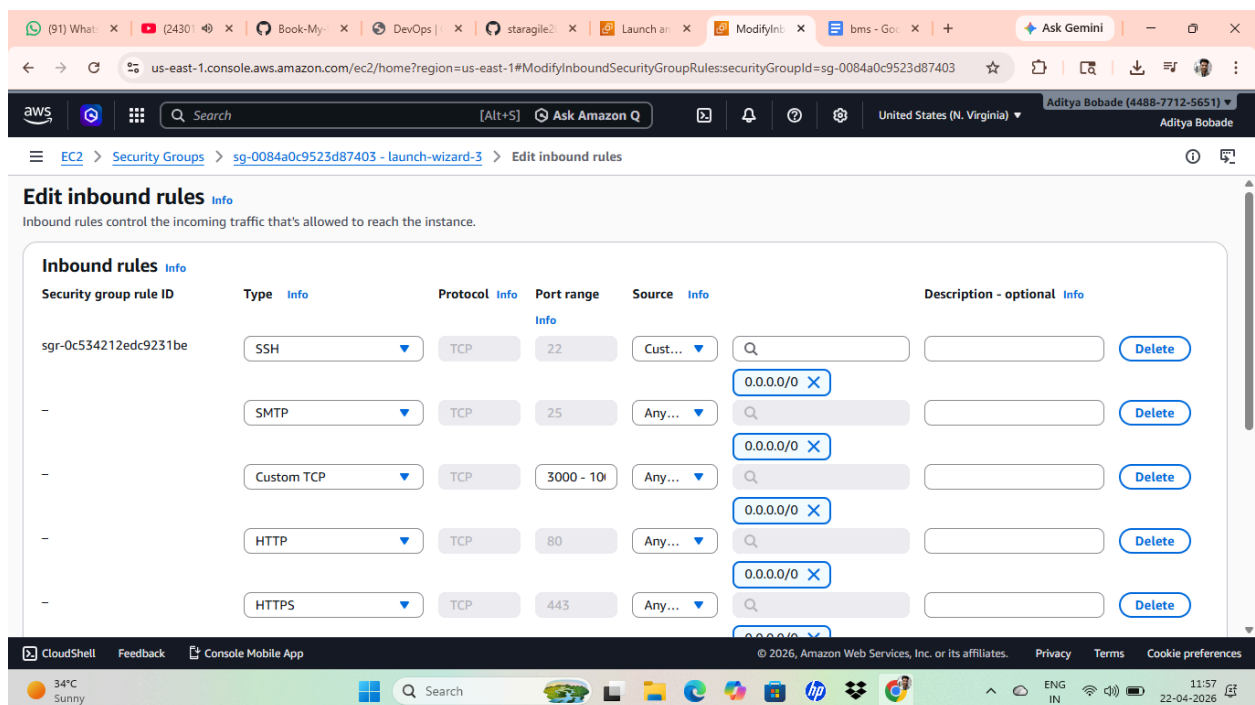
Name: Aditya Narendrapant Bobade

Mentor: Vikul sir

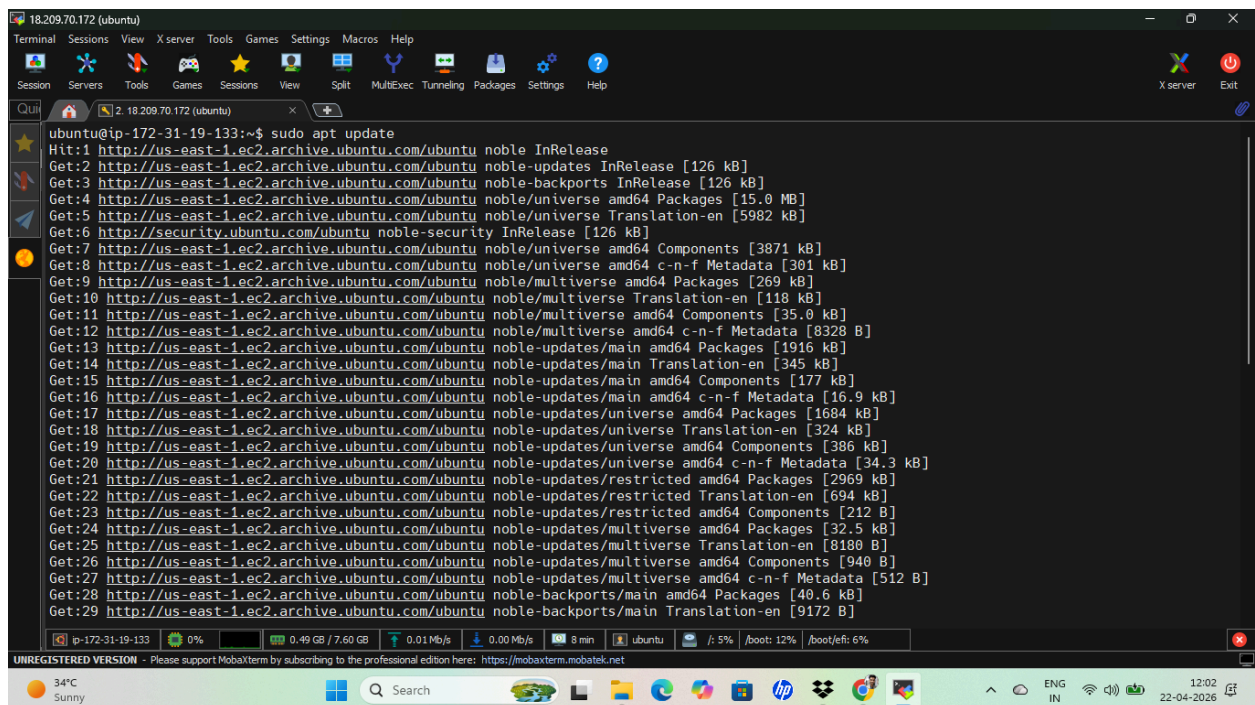
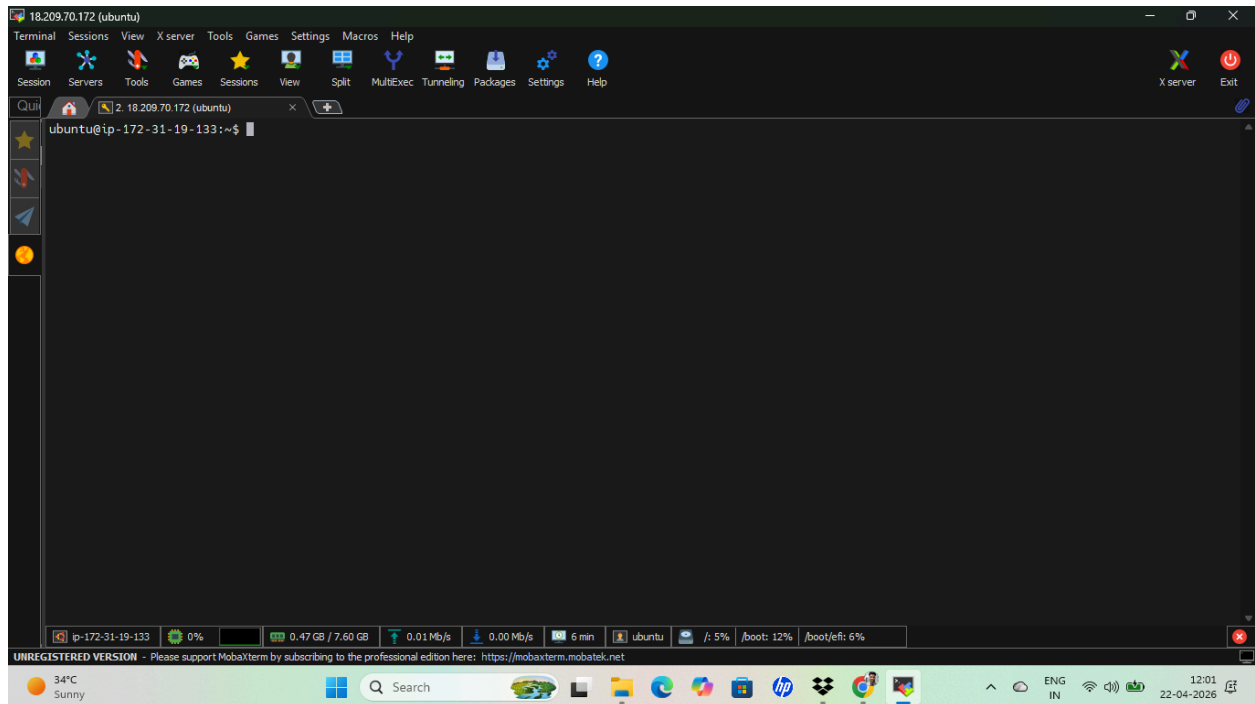
Step 1: launch virtual machine for the setup



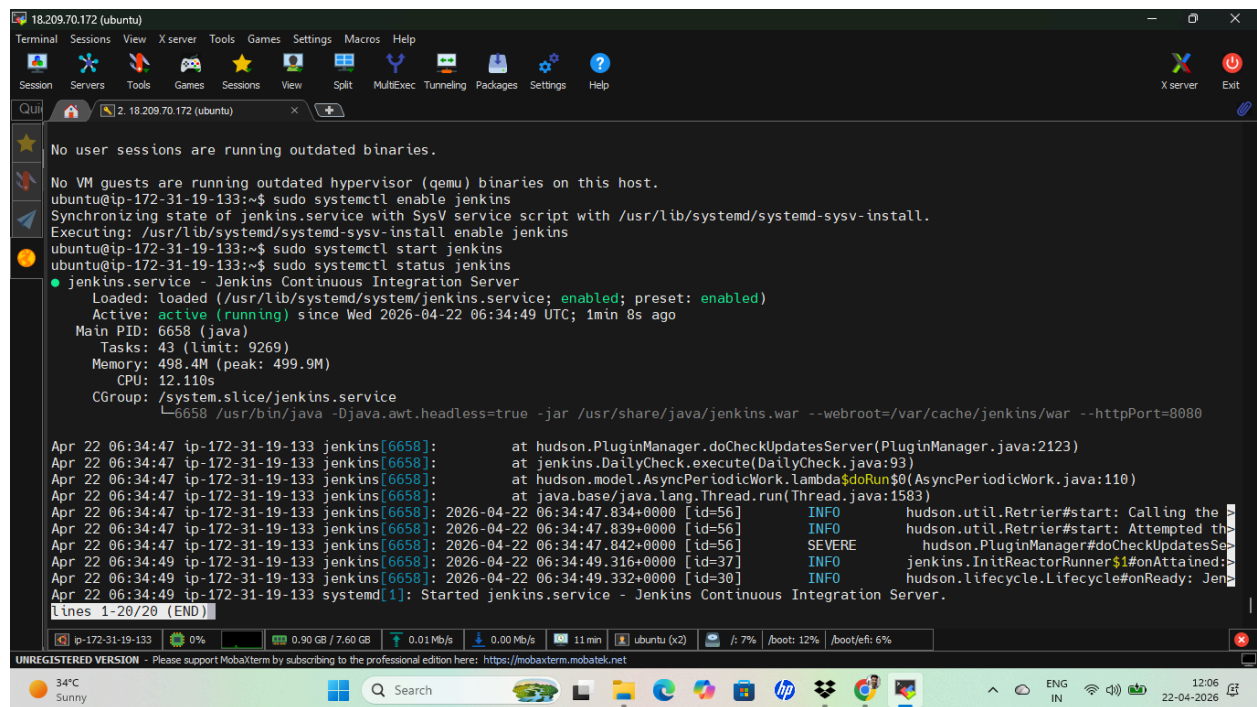
Step 2: Open below ports for the Security Group attached to the above VM SMTP, HTTP , CUSTOM TCP (3000-10000)



STEP 3: CONNECT THE VIRTUAL MACHINE THROUGH MOBAXTERM AND UPDATE IT



STEP 4: INSTALL JENKINS IN THE BMS SERVER



The screenshot shows a terminal window in MobaXterm with the following commands and output:

```
ubuntu@ip-172-31-19-133:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
ubuntu@ip-172-31-19-133:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-19-133:~$ sudo systemctl status jenkins
```

The status output for `jenkins.service` is as follows:

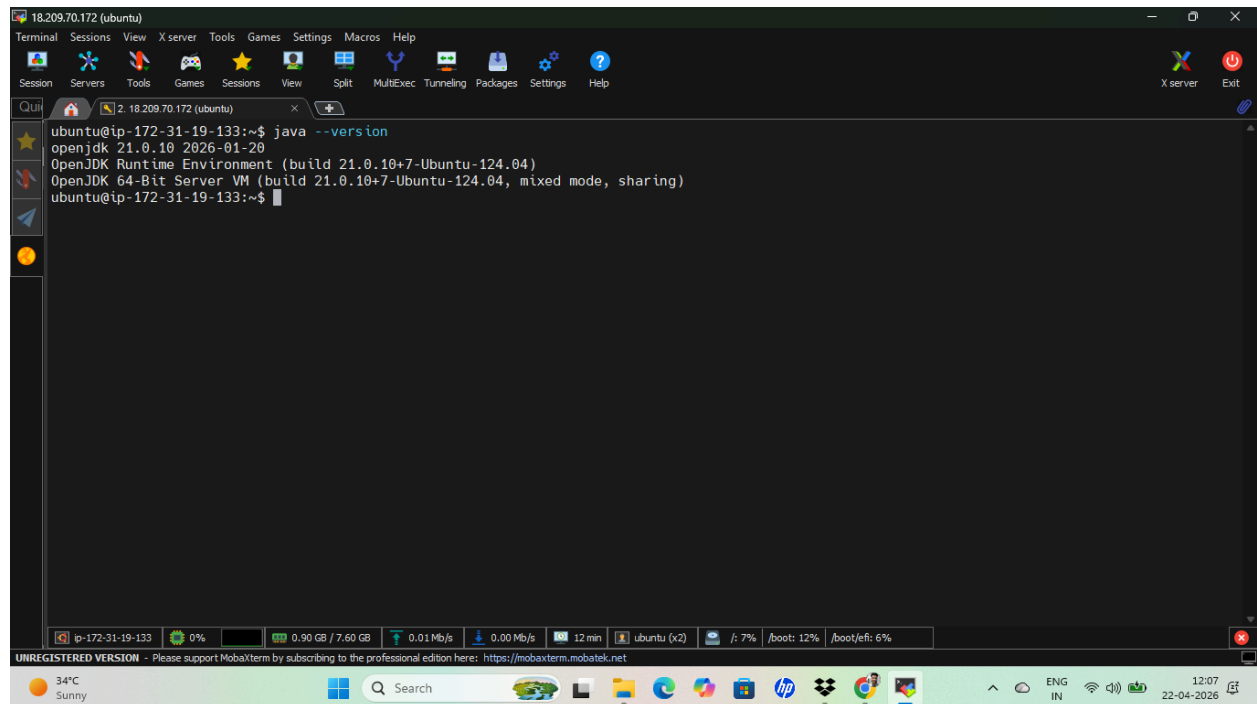
```
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 06:34:49 UTC; 1min 8s ago
     Main PID: 6658 (java)
       Tasks: 43 (limit: 9269)
      Memory: 498.4M (peak: 499.9M)
         CPU: 12.110s
        CGroup: /system.slice/jenkins.service
                └─6658 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
```

Below the status output, there are several log lines from the Jenkins process:

```
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: at hudson.PluginManager.doCheckUpdatesServer(PluginManager.java:2123)
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: at jenkins.DailyCheck.execute(DailyCheck.java:93)
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: at hudson.model.AsyncPeriodicWork.lambda$doRun$0(AsyncPeriodicWork.java:110)
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: at java.base/java.lang.Thread.run(Thread.java:1583)
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: 2026-04-22 06:34:47.834+0000 [id=56] INFO hudson.util.Retrier#start: Calling the
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: 2026-04-22 06:34:47.839+0000 [id=56] INFO hudson.util.Retrier#start: Attempted the
Apr 22 06:34:47 ip-172-31-19-133 jenkins[6658]: 2026-04-22 06:34:47.842+0000 [id=56] SEVERE hudson.PluginManager#doCheckUpdatesSe
Apr 22 06:34:49 ip-172-31-19-133 jenkins[6658]: 2026-04-22 06:34:49.316+0000 [id=37] INFO jenkins.InitReactorRunner$1#onAttained:
Apr 22 06:34:49 ip-172-31-19-133 jenkins[6658]: 2026-04-22 06:34:49.332+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady: Jen
Apr 22 06:34:49 ip-172-31-19-133 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
```

The terminal window also shows system resource usage at the bottom: 0.90 GB / 7.60 GB memory, 0.01 Mb/s upload, 0.00 Mb/s download, and 11 min runtime.

STEP 5: INSTALL JAVA IN THE SERVER

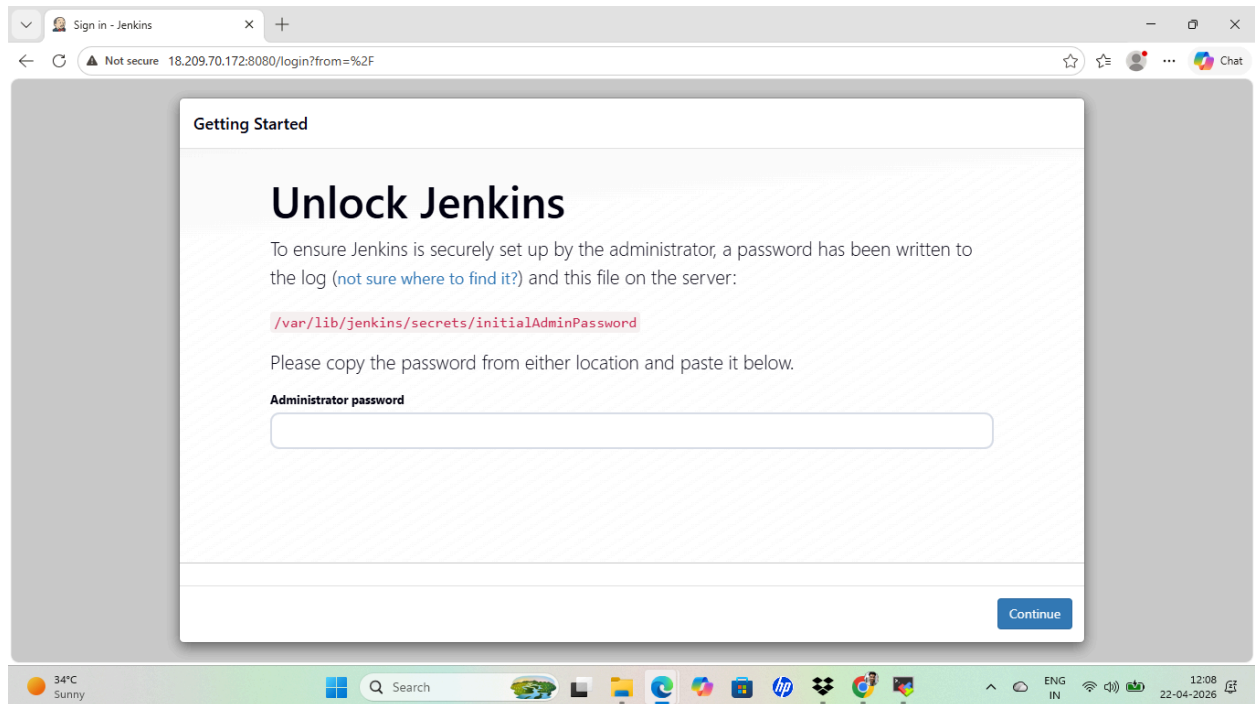


The screenshot shows a terminal window in MobaXterm with the following commands and output:

```
ubuntu@ip-172-31-19-133:~$ java --version
openjdk 21.0.10 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
ubuntu@ip-172-31-19-133:~$
```

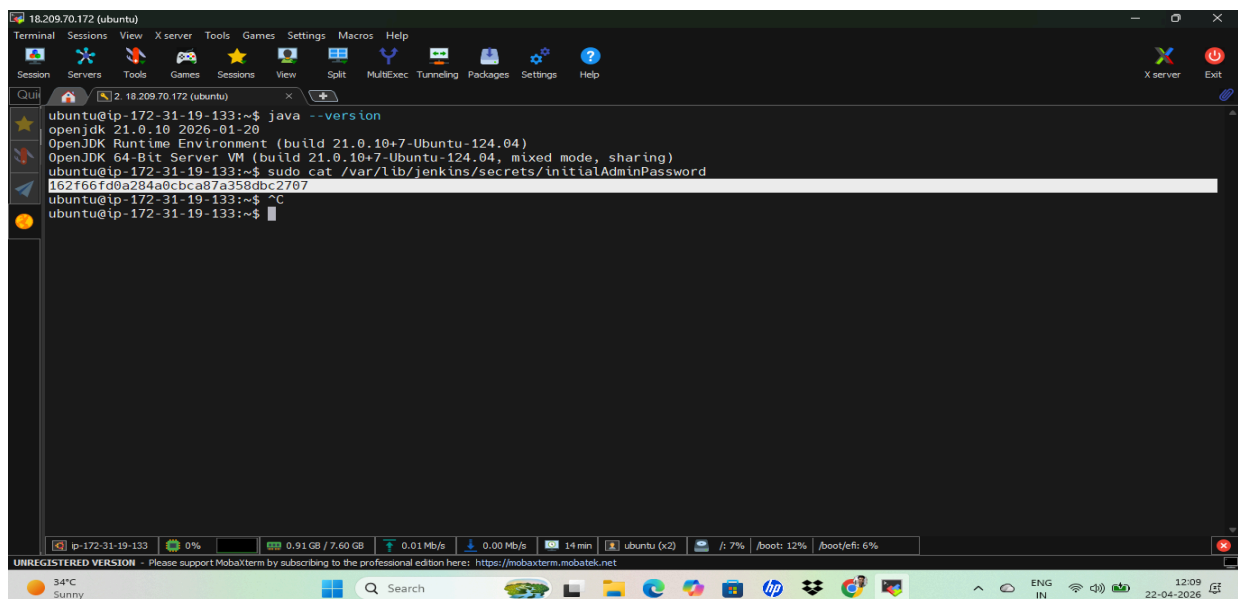
The terminal window also shows system resource usage at the bottom: 0.90 GB / 7.60 GB memory, 0.01 Mb/s upload, 0.00 Mb/s download, and 12 min runtime.

STEP 6: ACCESS JENKINS IN THE BROWSER USING PRIVATE IP AND JENKINS PORT NO(8080)

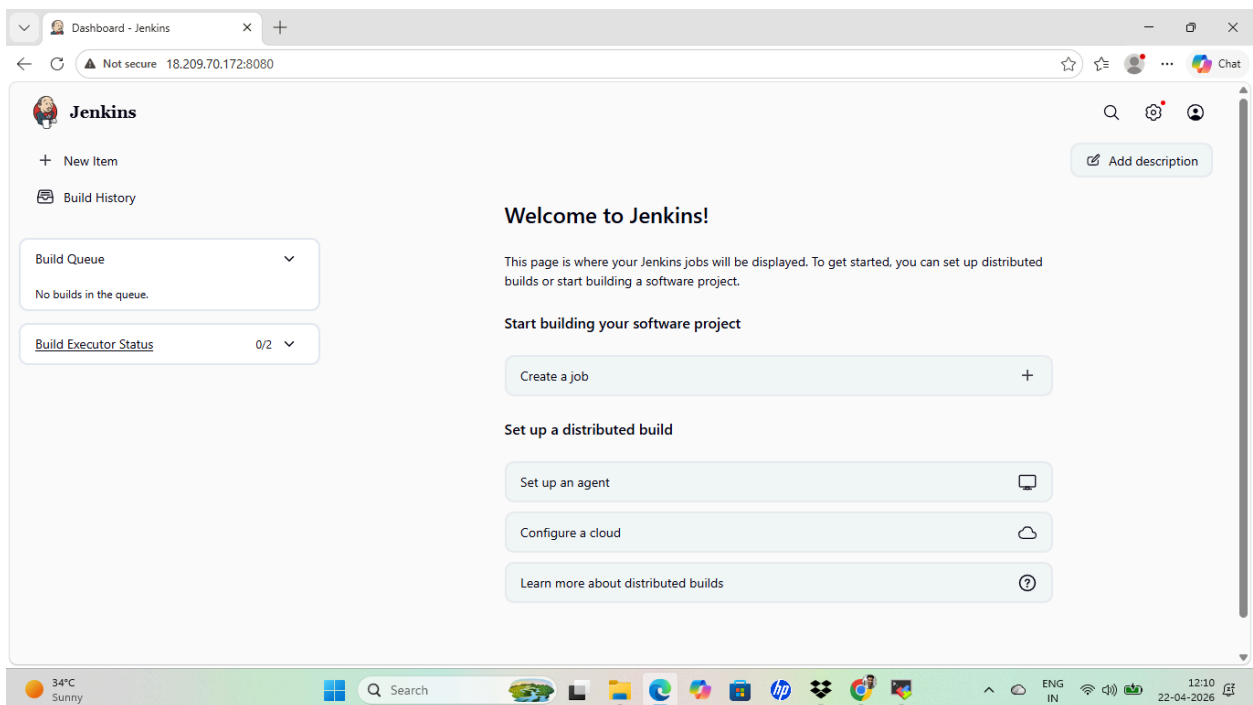
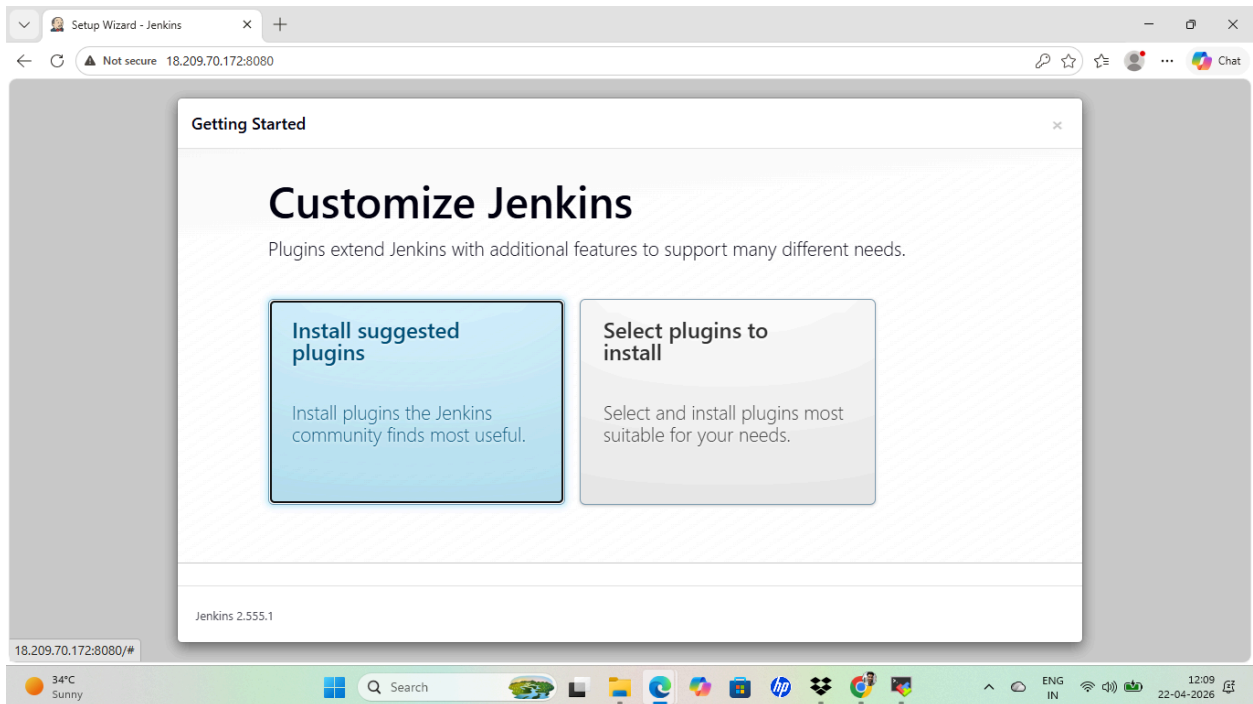


COPY PASSWORD FROM THIS PATH

`/var/lib/jenkins/secrets/initialAdminPassword`



STEP 7: INSTALL SUGGESTED PLUGINS FOR JENKINS



STEP 8: INSTALL DOCKER IN SERVER

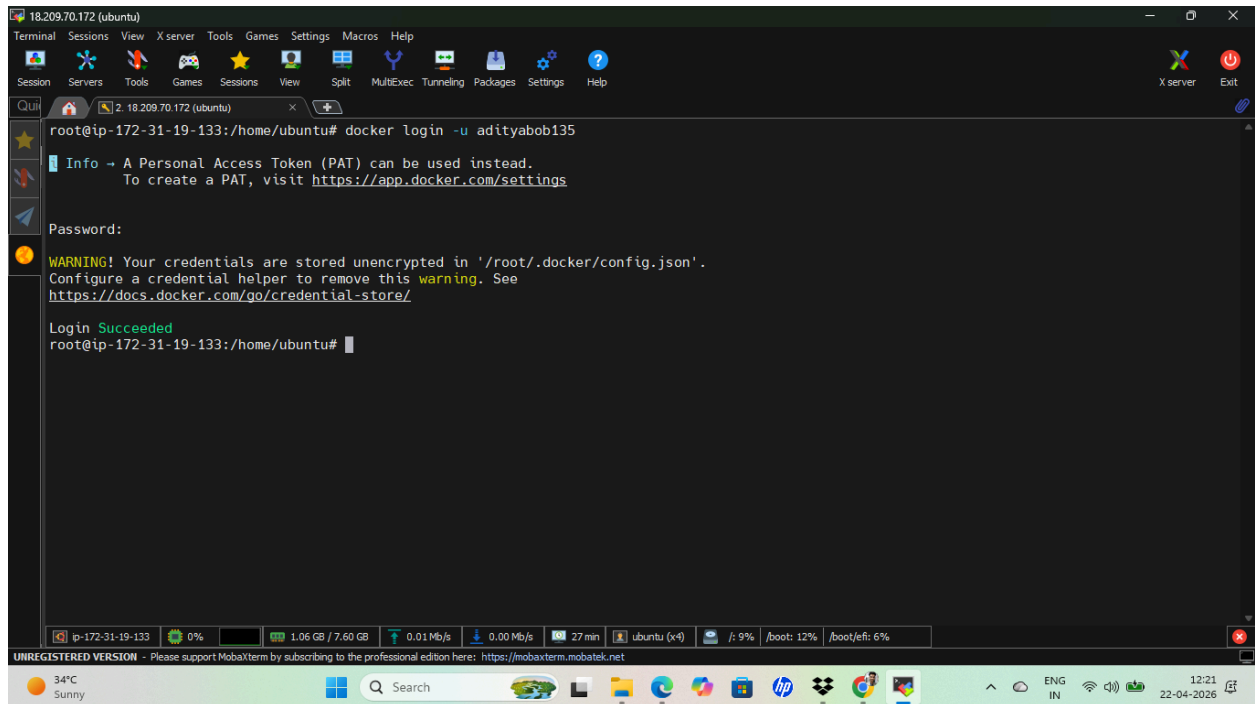
```
18.209.70.172 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quit 2. 18.209.70.172 (ubuntu)
ubuntu@ip-172-31-19-133:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 06:42:32 UTC; 8s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 11190 (dockerd)
      Tasks: 9
     Memory: 27.0M (peak: 27.1M)
        CPU: 199ms
   CGroup: /system.slice/docker.service
           └─11190 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.301866493Z" level=info msg="Restoring containers: start."
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.328784381Z" level=info msg="Deleting nftables IPv4 rules" error="exit
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.333999123Z" level=info msg="Deleting nftables IPv6 rules" error="exit
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.513031401Z" level=info msg="Loading containers: done."
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.518733998Z" level=info msg="Docker daemon" commit=6c91b92 containerd=
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.518866139Z" level=info msg="Initializing buildkit"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.547255546Z" level=info msg="Completed buildkit initialization"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.550412984Z" level=info msg="Daemon has completed initialization"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.550443364Z" level=info msg="API listen on /run/docker.sock"
Apr 22 06:42:32 ip-172-31-19-133 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

```
18.209.70.172 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quit 2. 18.209.70.172 (ubuntu)
ubuntu@ip-172-31-19-133:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 06:42:32 UTC; 8s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 11190 (dockerd)
      Tasks: 9
     Memory: 27.0M (peak: 27.1M)
        CPU: 199ms
   CGroup: /system.slice/docker.service
           └─11190 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.301866493Z" level=info msg="Restoring containers: start."
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.328784381Z" level=info msg="Deleting nftables IPv4 rules" error="exit
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.333999123Z" level=info msg="Deleting nftables IPv6 rules" error="exit
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.513031401Z" level=info msg="Loading containers: done."
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.518733998Z" level=info msg="Docker daemon" commit=6c91b92 containerd=
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.518866139Z" level=info msg="Initializing buildkit"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.547255546Z" level=info msg="Completed buildkit initialization"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.550412984Z" level=info msg="Daemon has completed initialization"
Apr 22 06:42:32 ip-172-31-19-133 dockerd[11190]: time="2026-04-22T06:42:32.550443364Z" level=info msg="API listen on /run/docker.sock"
Apr 22 06:42:32 ip-172-31-19-133 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
[2]+  Stopped                  sudo systemctl status docker
ubuntu@ip-172-31-19-133:~$ sudo systemctl start docker
ubuntu@ip-172-31-19-133:~$ docker --version
Docker version 29.4.1, build 055a478
ubuntu@ip-172-31-19-133:~$
```

STEP 9 : LOGIN IN DOCKER HUB



The screenshot shows a terminal window titled "18.209.70.172 (ubuntu)". The terminal displays the command `docker login -u adityabob135`. An information message states: "Info → A Personal Access Token (PAT) can be used instead. To create a PAT, visit <https://app.docker.com/settings>". A password prompt is shown, followed by a warning: "WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'. Configure a credential helper to remove this warning. See <https://docs.docker.com/go/credential-store/>". The login is successful, and the prompt returns to `root@ip-172-31-19-133:/home/ubuntu#`. The terminal interface includes a menu bar (Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, Help) and a toolbar with icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, and Help. The status bar at the bottom shows system information: 34°C Sunny, search bar, and system tray with date 22-04-2026 and time 12:21.

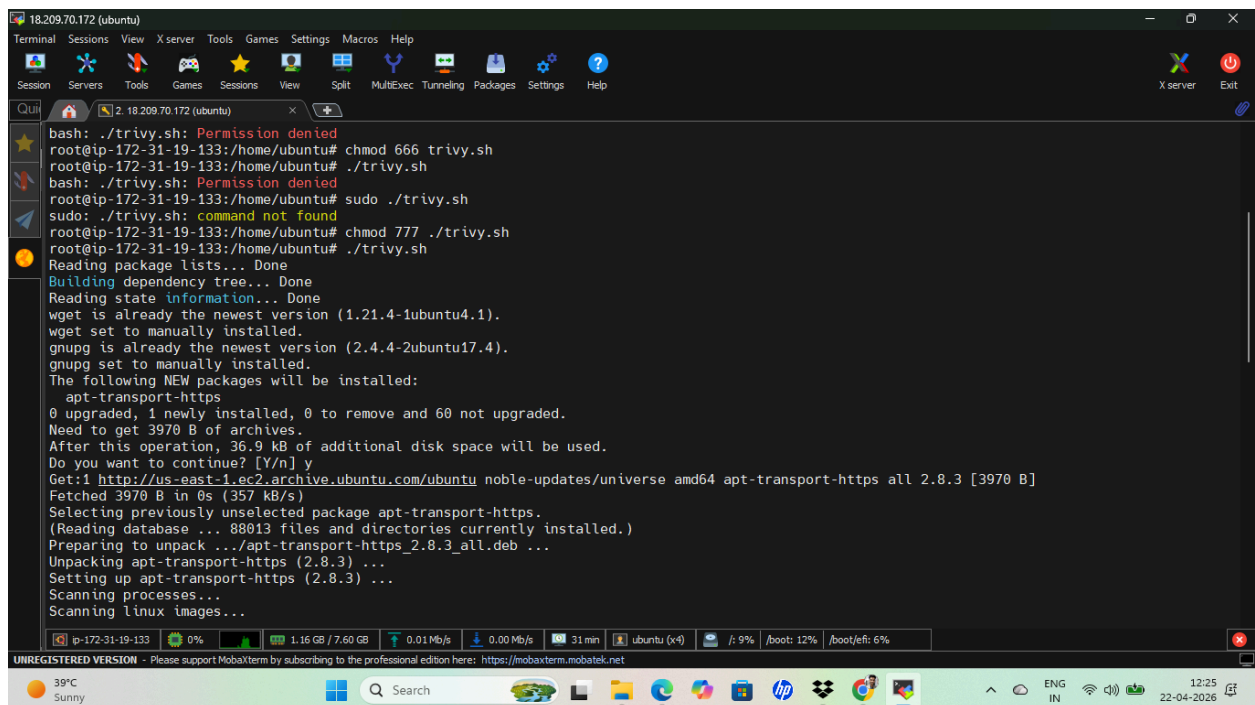
```
root@ip-172-31-19-133:/home/ubuntu# docker login -u adityabob135

Info → A Personal Access Token (PAT) can be used instead.
        To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

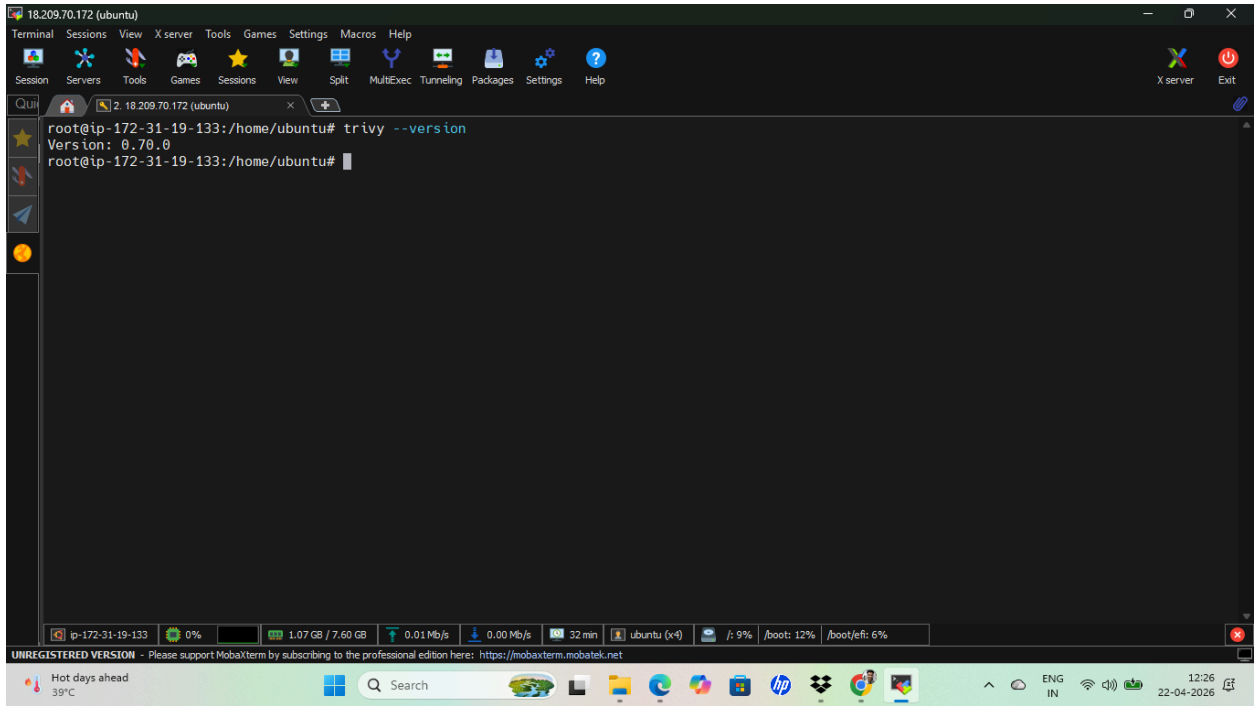
Login Succeeded
root@ip-172-31-19-133:/home/ubuntu#
```



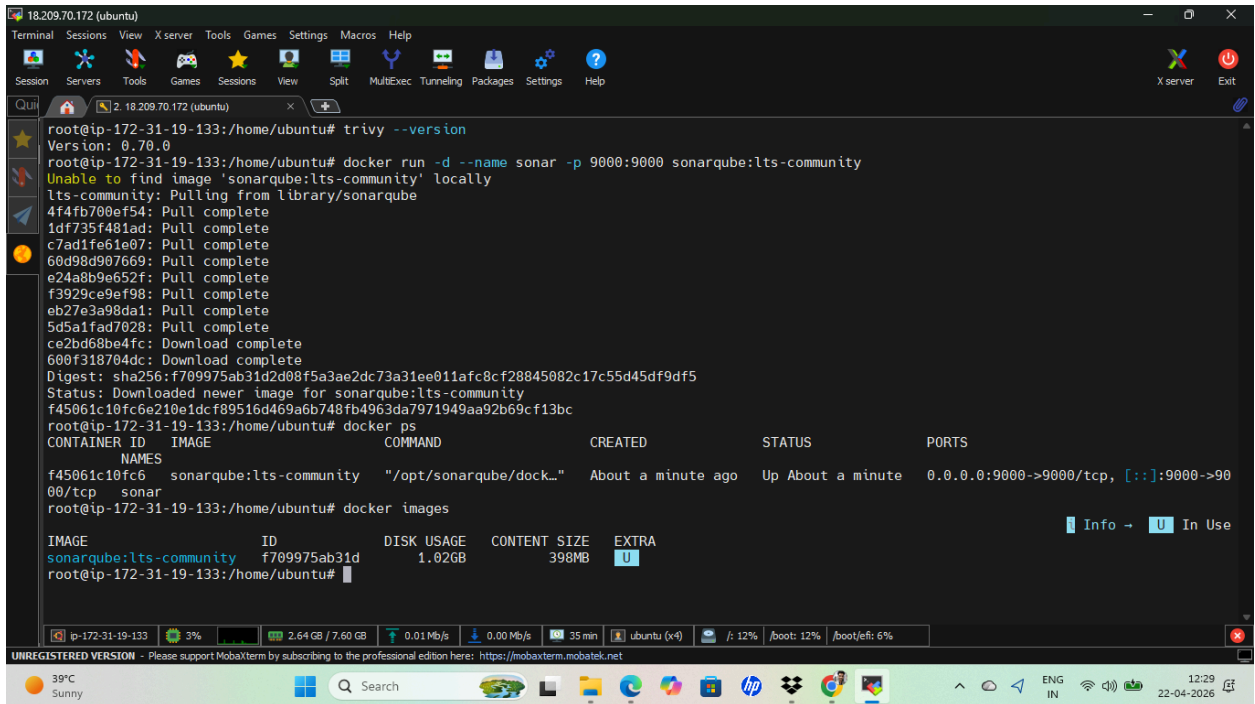
The screenshot shows a terminal window titled "18.209.70.172 (ubuntu)". The terminal displays the command `bash: ./trivy.sh: Permission denied`. The user then runs `chmod 666 trivy.sh`. The command is still denied, so the user runs `chmod 777 ./trivy.sh`. The user then runs `sudo ./trivy.sh`. The terminal shows the output of the script, which includes the following text: "Reading package lists... Done", "Building dependency tree... Done", "Reading state information... Done", "wget is already the newest version (1.21.4-1ubuntu4.1).", "wget set to manually installed.", "gnupg is already the newest version (2.4.4-2ubuntu17.4).", "gnupg set to manually installed.", "The following NEW packages will be installed:", "apt-get", "apt-get is already the newest version (1.21.4-1ubuntu4.1).", "Need to get 3970 B of archives.", "After this operation, 36.9 kB of additional disk space will be used.", "Do you want to continue? [Y/n] y", "Get:1 <http://us-east-1.ec2.archive.ubuntu.com/ubuntu> noble-updates/universe amd64 apt-get all 2.8.3 [3970 B]", "Fetched 3970 B in 0s (357 kB/s)", "Selecting previously unselected package apt-get.", "(Reading database ... 88013 files and directories currently installed.)", "Preparing to unpack .../apt-get_2.8.3_all.deb ...", "Unpacking apt-get (2.8.3) ...", "Setting up apt-get (2.8.3) ...", "Scanning processes...", "Scanning linux images...". The terminal interface includes a menu bar (Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, Help) and a toolbar with icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, and Help. The status bar at the bottom shows system information: 39°C Sunny, search bar, and system tray with date 22-04-2026 and time 12:25.

```
bash: ./trivy.sh: Permission denied
root@ip-172-31-19-133:/home/ubuntu# chmod 666 trivy.sh
root@ip-172-31-19-133:/home/ubuntu# ./trivy.sh
bash: ./trivy.sh: Permission denied
root@ip-172-31-19-133:/home/ubuntu# sudo ./trivy.sh
sudo: ./trivy.sh: command not found
root@ip-172-31-19-133:/home/ubuntu# chmod 777 ./trivy.sh
root@ip-172-31-19-133:/home/ubuntu# ./trivy.sh
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
The following NEW packages will be installed:
apt-get
0 upgraded, 1 newly installed, 0 to remove and 60 not upgraded.
Need to get 3970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-get all 2.8.3 [3970 B]
Fetched 3970 B in 0s (357 kB/s)
Selecting previously unselected package apt-get.
(Reading database ... 88013 files and directories currently installed.)
Preparing to unpack .../apt-get_2.8.3_all.deb ...
Unpacking apt-get (2.8.3) ...
Setting up apt-get (2.8.3) ...
Scanning processes...
Scanning linux images...
```

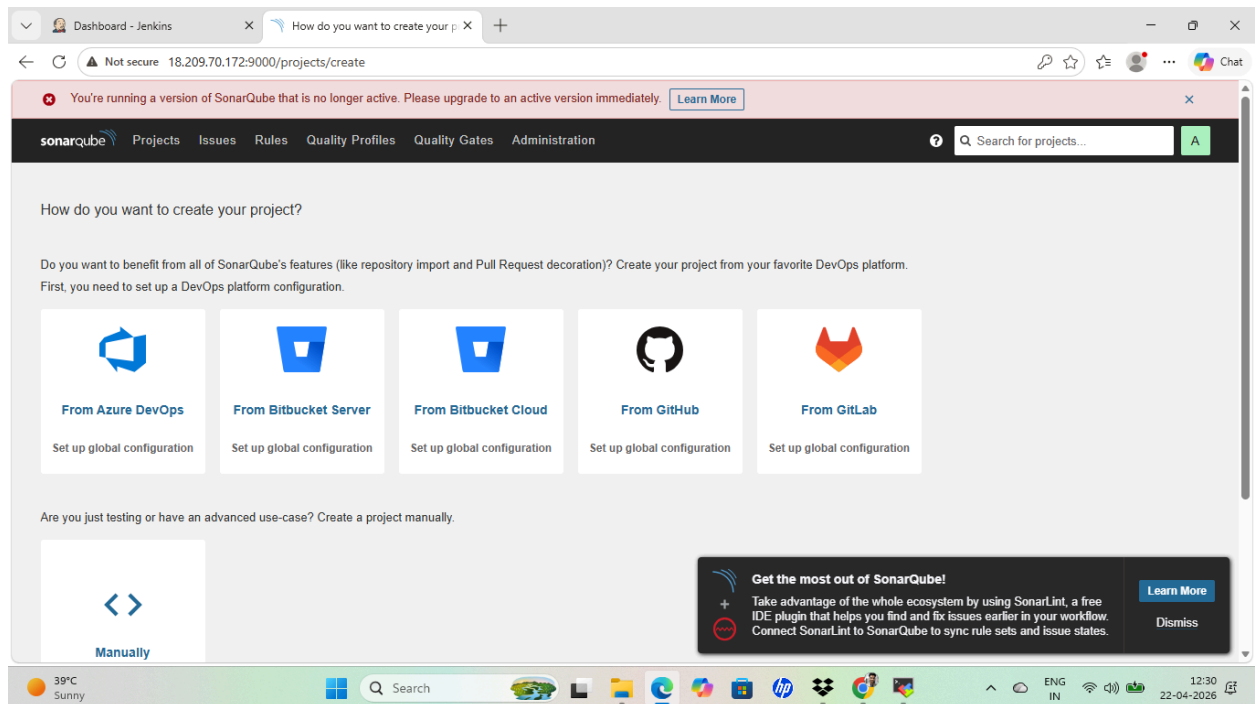
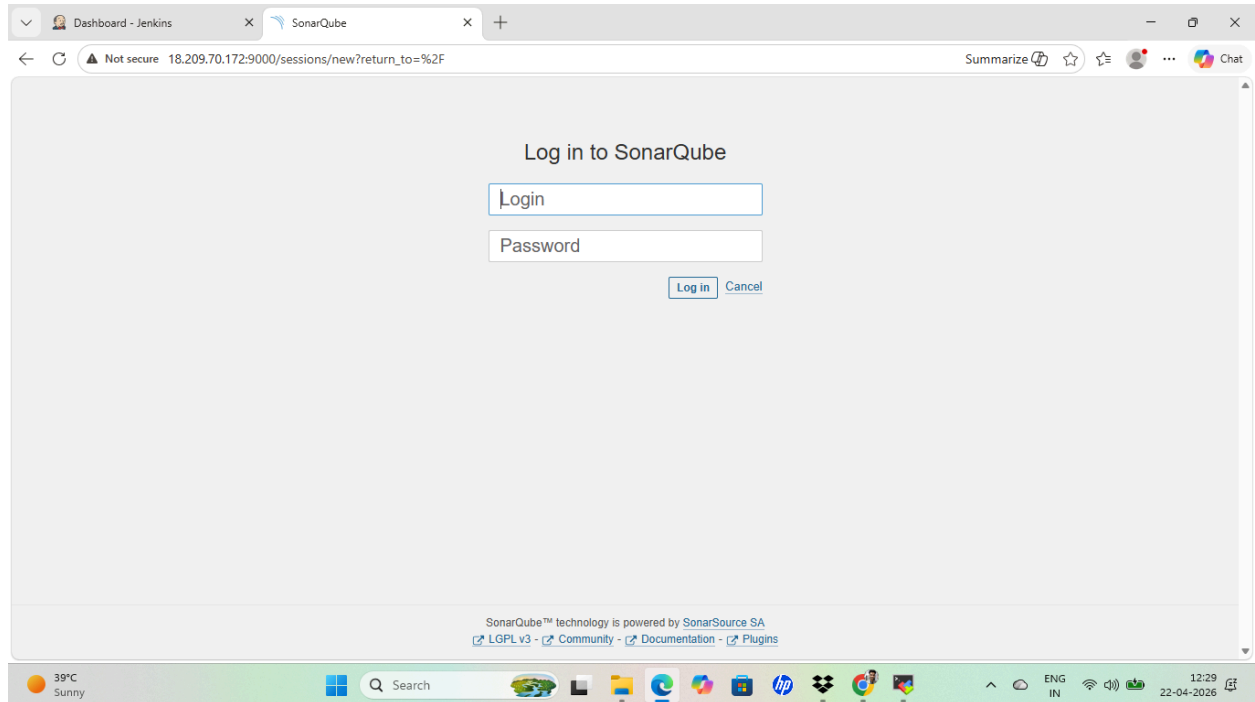
STEP 10 : INSTALL TRIVY



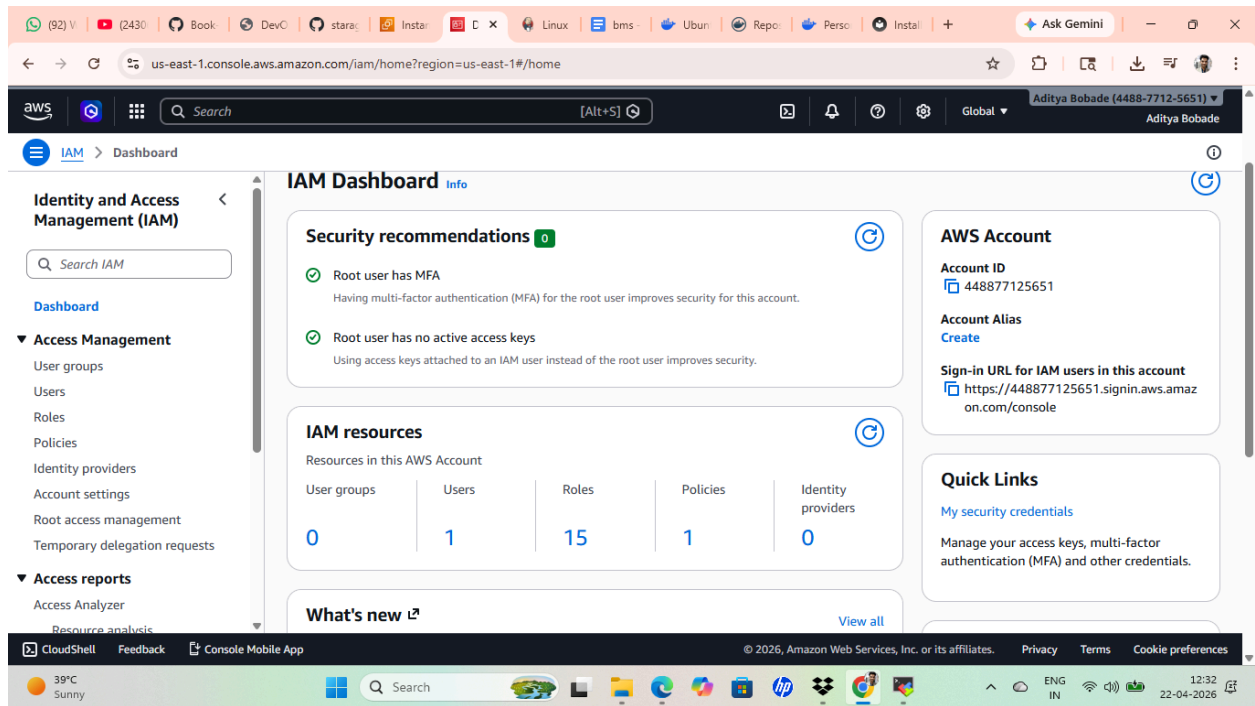
STEP 11: LAUNCH DOCKER CONTAINER FOR SONARQUBE



STEP 12: LODIN INTO SONARQUBE USING BROWSER USING PUBLIC IP AND SONARQUBE PORT NO(9000)

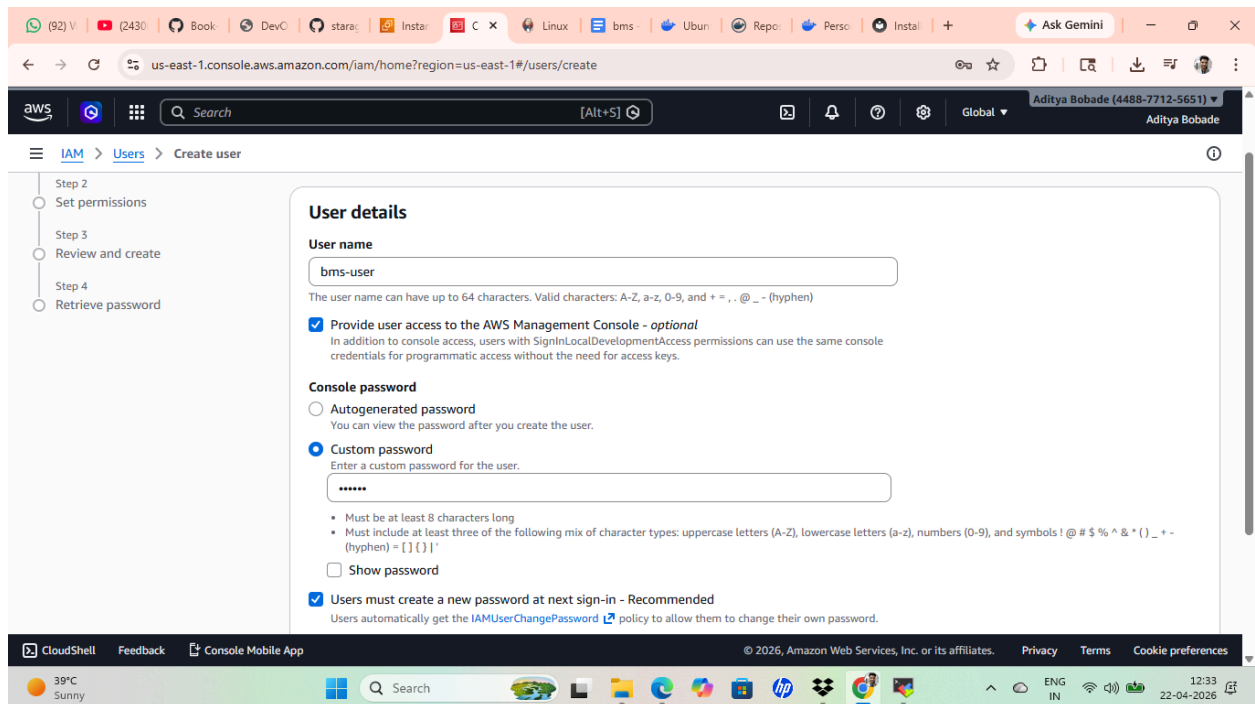


STEP 13: CREATE AN IAM USER



The screenshot shows the AWS IAM Dashboard in the us-east-1 region. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like 'Dashboard', 'Access Management', and 'Access reports'. The main content area is titled 'IAM Dashboard' and includes several sections: 'Security recommendations' showing two checks for MFA and access keys; 'IAM resources' with a table of counts for user groups (0), users (1), roles (15), policies (1), and identity providers (0); 'AWS Account' information including account ID and alias; and 'Quick Links' for security credentials. A 'What's new' section is also visible at the bottom.

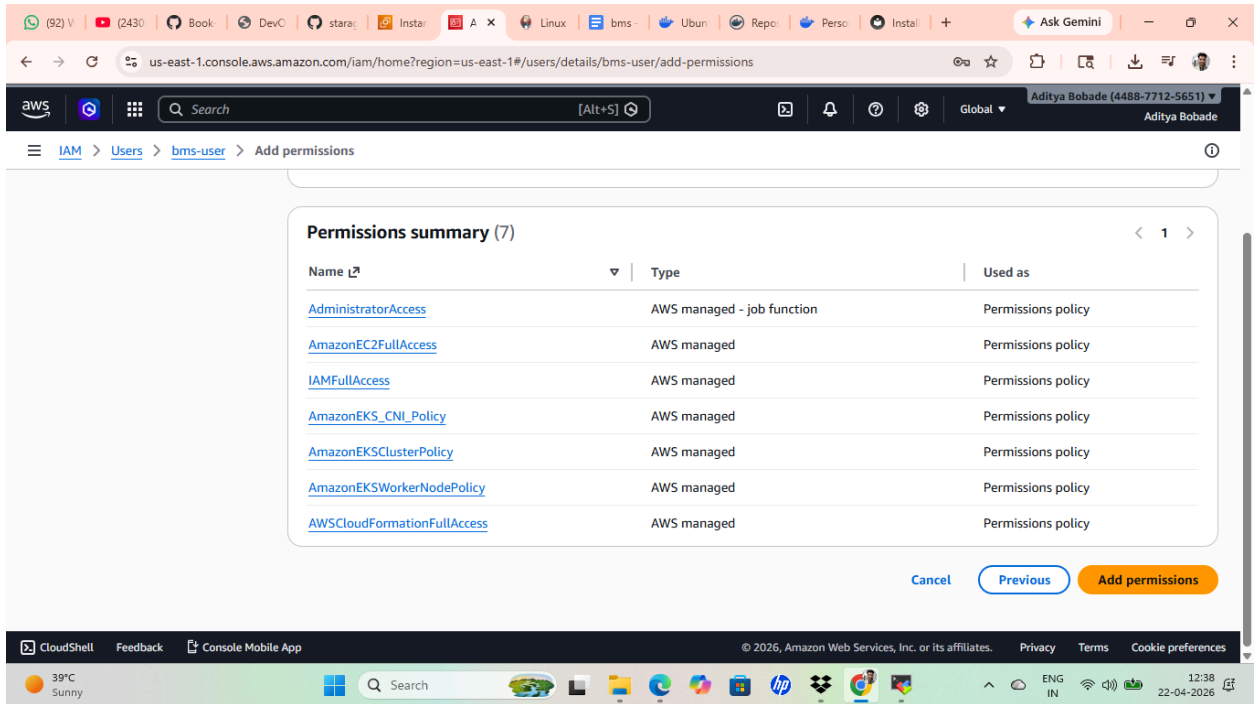
IAM resources				
User groups	Users	Roles	Policies	Identity providers
0	1	15	1	0



The screenshot shows the 'Create user' wizard in the AWS IAM console. The left sidebar indicates the current step is 'Step 2: Set permissions'. The main content area is titled 'User details' and contains the following sections:

- User name:** A text field containing 'bms-user'. Below it, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, ., @, _ (hyphen)'.
- Provide user access to the AWS Management Console - optional:** A checked checkbox. Below it, a note states: 'In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.'
- Console password:** A section with two options: 'Autogenerated password' (unselected) and 'Custom password' (selected). The 'Custom password' option includes a text field with masked characters and a list of requirements: 'Must be at least 8 characters long' and 'Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ \$ % ^ & * () _ + - (hyphen) = [] { } | ' ' '. There is also a 'Show password' checkbox.
- Users must create a new password at next sign-in - Recommended:** A checked checkbox. Below it, a note states: 'Users automatically get the IAMUserChangePassword policy to allow them to change their own password.'

STEP 14: GIVE REQUIRED PERMISSIONS TO THE USERS



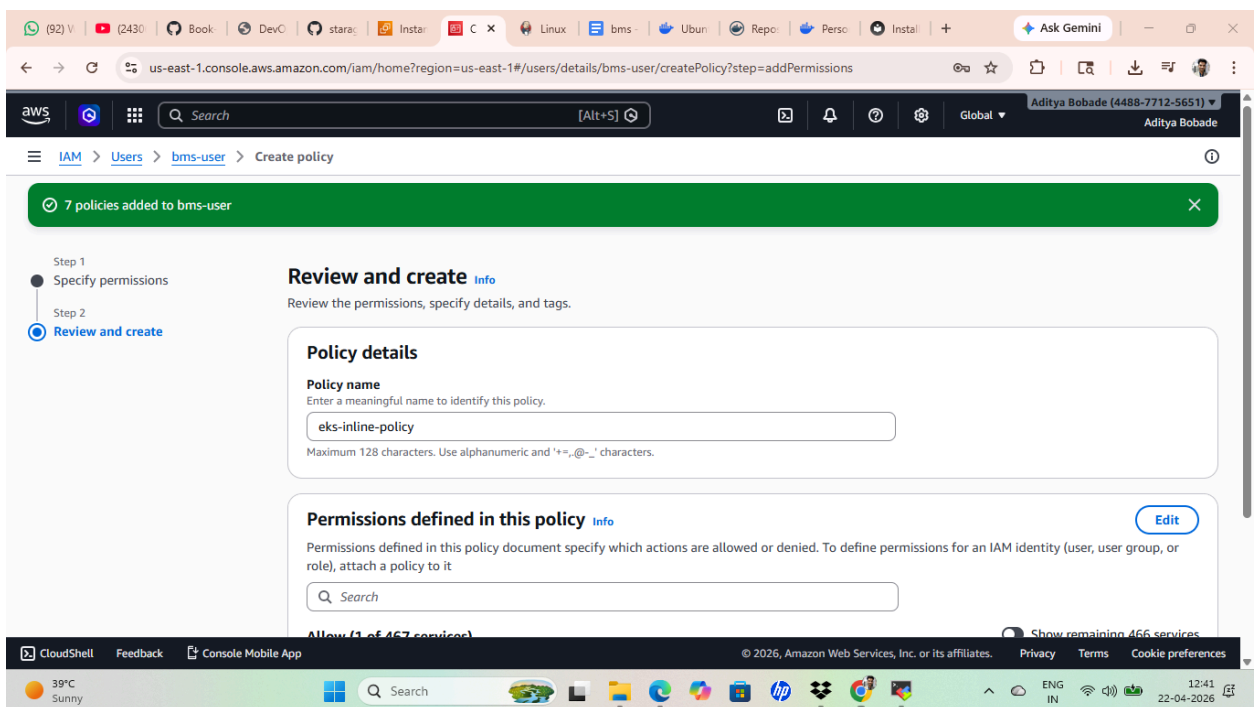
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/bms-user/add-permissions

Permissions summary (7)

Name	Type	Used as
AdministratorAccess	AWS managed - job function	Permissions policy
AmazonEC2FullAccess	AWS managed	Permissions policy
IAMFullAccess	AWS managed	Permissions policy
AmazonEKS_CNI_Policy	AWS managed	Permissions policy
AmazonEKSClusterPolicy	AWS managed	Permissions policy
AmazonEKSWorkerNodePolicy	AWS managed	Permissions policy
AWSCloudFormationFullAccess	AWS managed	Permissions policy

Cancel Previous Add permissions

STEP 15: CREATE AN INLINE POLICY FOR EKS CLUSTER



us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/bms-user/createPolicy?step=addPermissions

7 policies added to bms-user

Step 1 Specify permissions
Step 2 Review and create

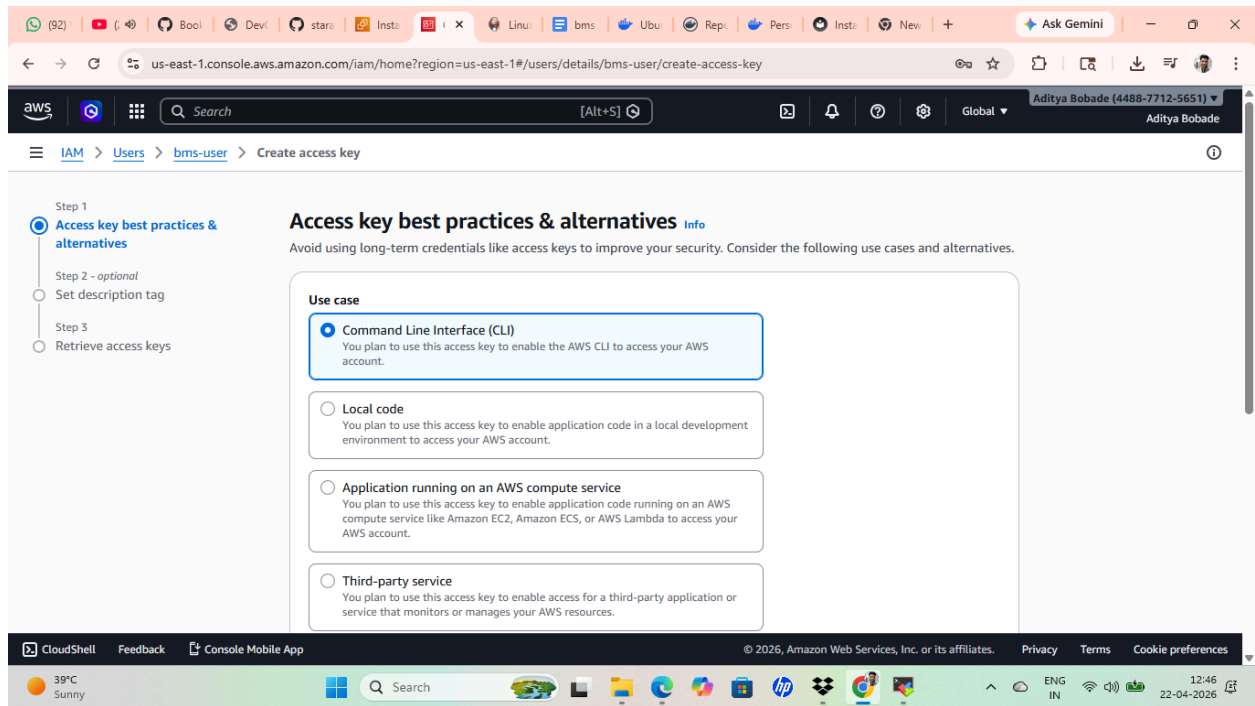
Review and create Info

Review the permissions, specify details, and tags.

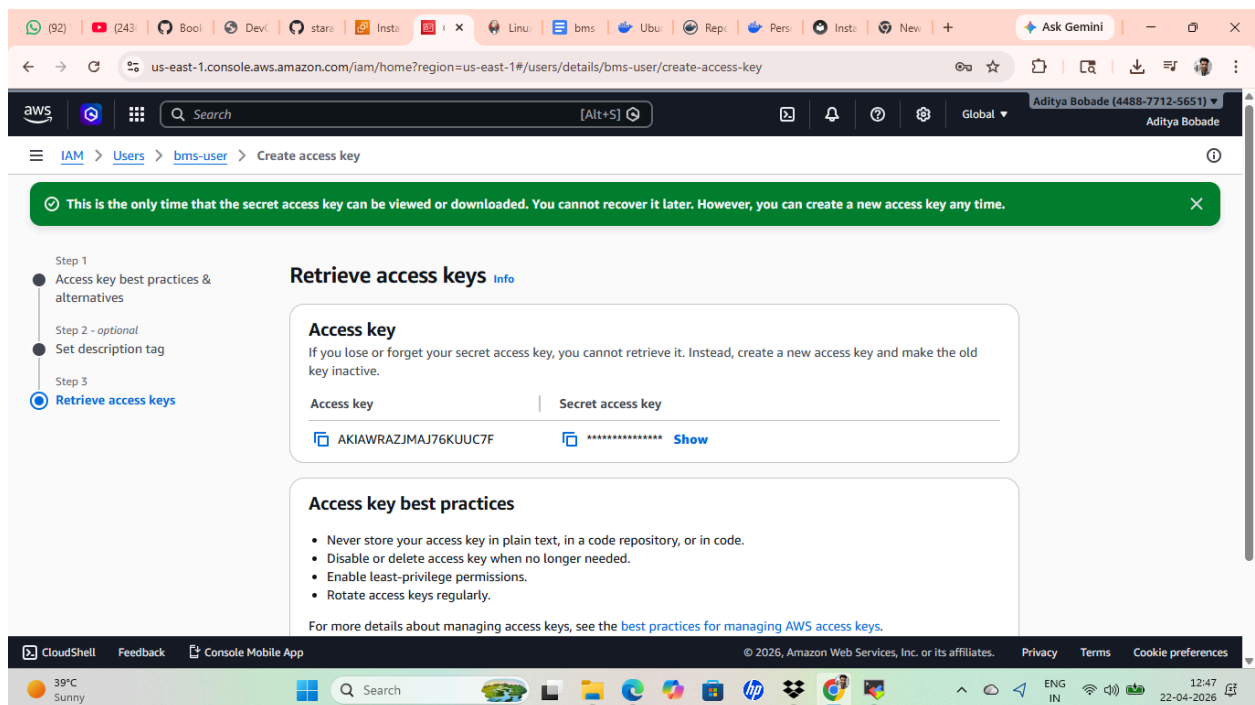
Policy details

Policy name
Enter a meaningful name to identify this policy.
eks-inline-policy
Maximum 128 characters. Use alphanumeric and '+', '=', '@', '-', '_' characters.

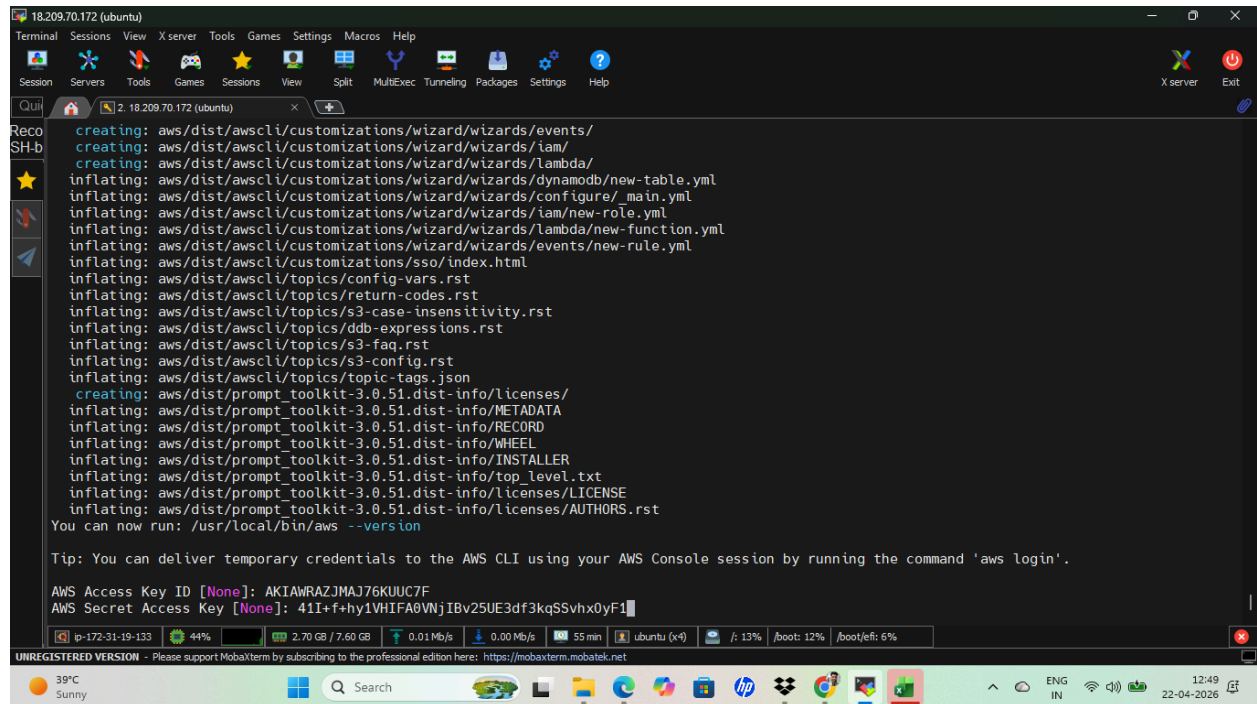
Permissions defined in this policy Info Edit
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.
Search
Allow (1 of 467 services)



STEP 16: CREATE ACCESS KEYS FOR CONFIGURING AWS CLI IN BROWSER



STEP 17: DOWNLOAD AWS CLI AND CONFIGURE IT USING ACCESS KEY ID AND ACCESS KEY



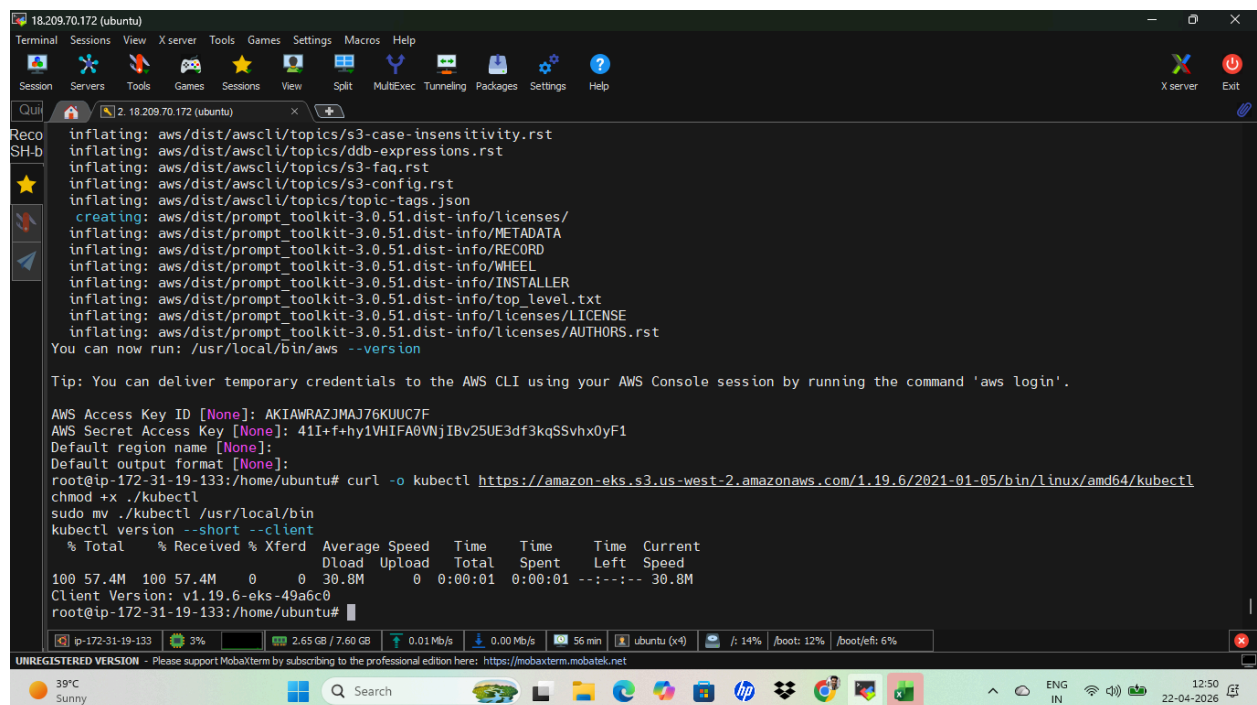
```
18.209.70.172 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Qu... 2. 18.209.70.172 (ubuntu)
Reco
SH-b
★
inflatng: aws/dist/awscli/customizations/wizard/wizards/events/
creating: aws/dist/awscli/customizations/wizard/wizards/iam/
creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
inflatng: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
inflatng: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
inflatng: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
inflatng: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
inflatng: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
inflatng: aws/dist/awscli/customizations/sso/index.html
inflatng: aws/dist/awscli/topics/config-vars.rst
inflatng: aws/dist/awscli/topics/return-codes.rst
inflatng: aws/dist/awscli/topics/s3-case-insensitivity.rst
inflatng: aws/dist/awscli/topics/db-expressions.rst
inflatng: aws/dist/awscli/topics/s3-faq.rst
inflatng: aws/dist/awscli/topics/s3-config.rst
inflatng: aws/dist/awscli/topics/topic-tags.json
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
You can now run: /usr/local/bin/aws --version

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAWRAZJMAJ76KUUC7F
AWS Secret Access Key [None]: 41i+f+hy1VHIFA0VNjIBv25UE3df3kqSSvhx0yF1

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

STEP 18: DOWNLOAD KUBECTL IN INSTANCE



```
18.209.70.172 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Qu... 2. 18.209.70.172 (ubuntu)
Reco
SH-b
★
inflatng: aws/dist/awscli/topics/s3-case-insensitivity.rst
inflatng: aws/dist/awscli/topics/db-expressions.rst
inflatng: aws/dist/awscli/topics/s3-faq.rst
inflatng: aws/dist/awscli/topics/s3-config.rst
inflatng: aws/dist/awscli/topics/topic-tags.json
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflatng: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
You can now run: /usr/local/bin/aws --version

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAWRAZJMAJ76KUUC7F
AWS Secret Access Key [None]: 41i+f+hy1VHIFA0VNjIBv25UE3df3kqSSvhx0yF1
Default region name [None]:
Default output format [None]:
root@ip-172-31-19-133:/home/ubuntu# curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin
kubectl version --short --client
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 57.4M 100 57.4M 0 0 30.8M 0:00:01 0:00:01 0:00:01 30.8M
Client Version: v1.19.6-eks-49a6c0
root@ip-172-31-19-133:/home/ubuntu#

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```

STEP 19: CREATE EKS CLUSTER IN MULTI AVAILABILITY ZONE

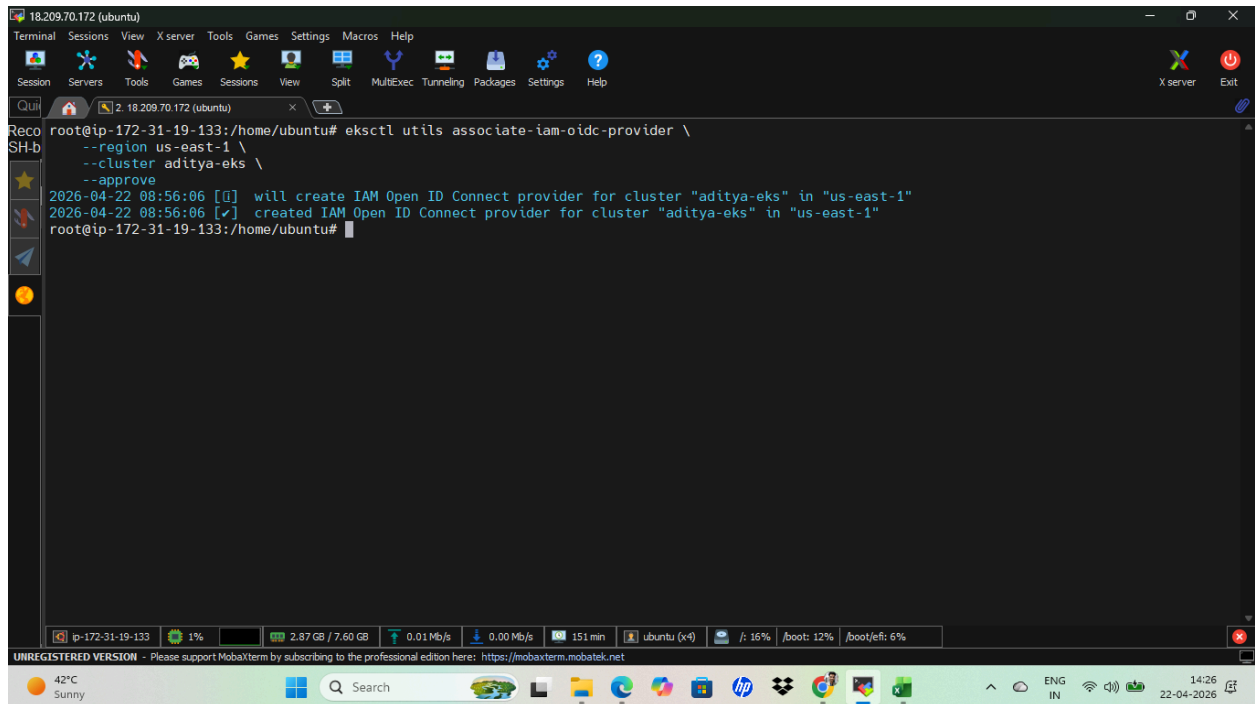
```
eksctl create cluster --name=aditya-eks \  
--region=us-east-1 \  
--zones=us-east-1a,us-east-1b \  
--version=1.30 \  
--without-nodegroup
```

The screenshot displays a terminal window on an Ubuntu system (IP: 18.209.70.172) where the command `eksctl create cluster --name=aditya-eks --region=us-east-1 --zones=us-east-1a,us-east-1b --version=1.30 --without-nodegroup` was executed. The terminal output shows the following steps:

- building cluster stack "eksctl-aditya-eks-cluster"
- deploying stack "eksctl-aditya-eks-cluster"
- waiting for CloudFormation stack "eksctl-aditya-eks-cluster"
- recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update addon'
- creating addon: vpc-cni
- successfully created addon: vpc-cni
- creating addon: kube-proxy
- successfully created addon: kube-proxy
- creating addon: coredns
- successfully created addon: coredns
- waiting for the control plane to become ready
- saved kubeconfig as "/root/.kube/config"
- no tasks
- all EKS cluster resources for "aditya-eks" have been created
- creating addon: metrics-server
- successfully created addon: metrics-server
- kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
- EKS cluster "aditya-eks" in "us-east-1" region is ready

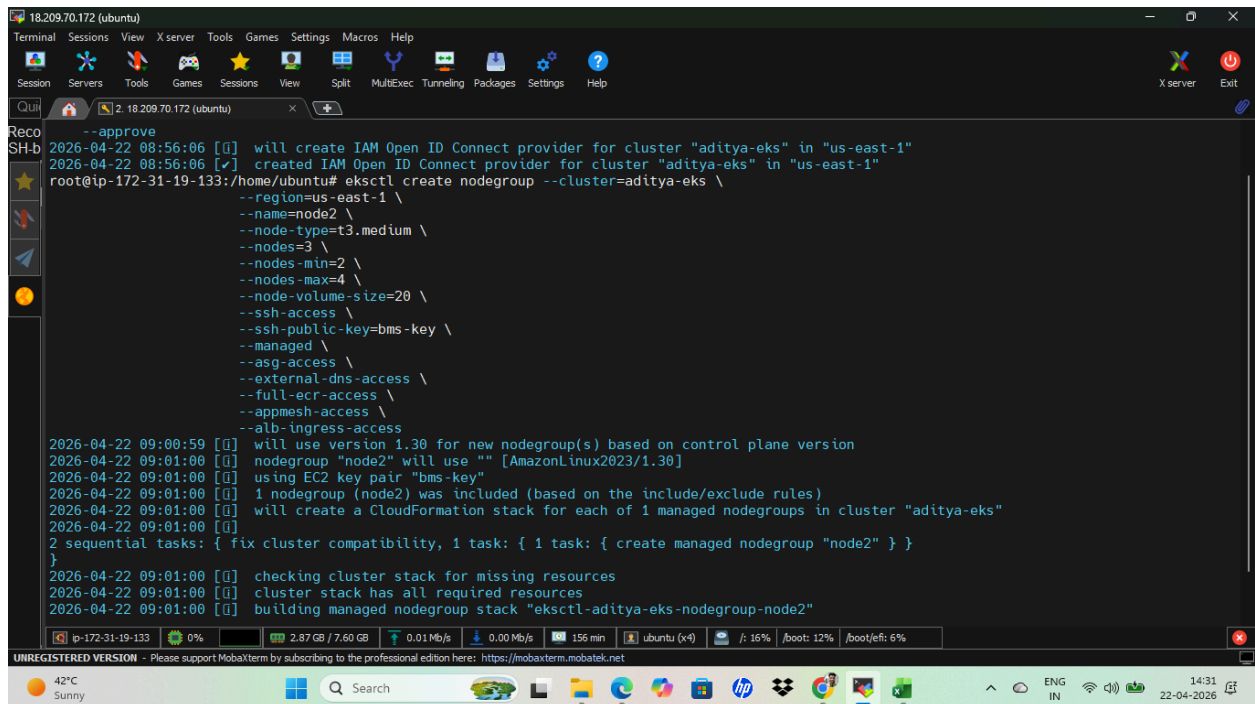
The AWS console shows the cluster 'aditya-eks' in the 'Active' state, with a Kubernetes version of 1.30 and an 'Extended support until July 23, 2026' status. The console also displays the 'Settings' and 'Related services' sections.

STEP 20: CREATE IAM OIDC PROVIDER



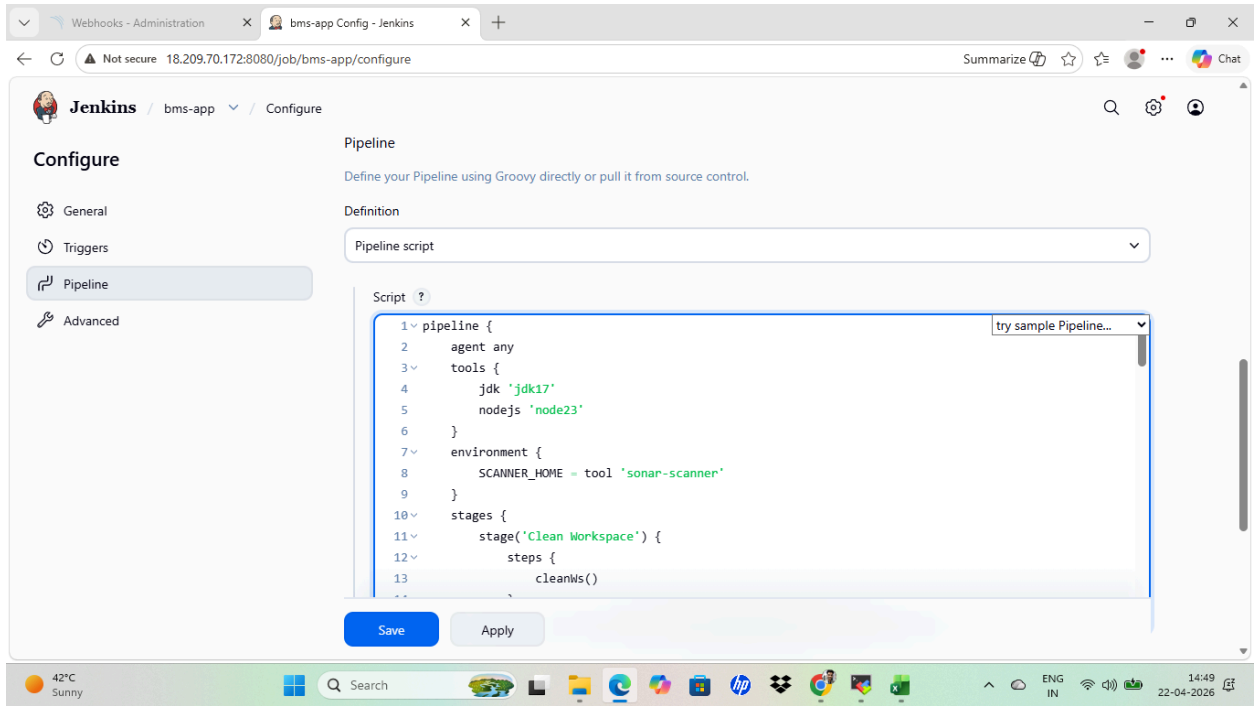
```
root@ip-172-31-19-133:/home/ubuntu# eksctl utils associate-iam-oidc-provider \
--region us-east-1 \
--cluster aditya-eks \
--approve
2026-04-22 08:56:06 [i] will create IAM Open ID Connect provider for cluster "aditya-eks" in "us-east-1"
2026-04-22 08:56:06 [✓] created IAM Open ID Connect provider for cluster "aditya-eks" in "us-east-1"
root@ip-172-31-19-133:/home/ubuntu#
```

STEP 21: CREATE NODE GROUP FOR EKS CLUSTER



```
root@ip-172-31-19-133:/home/ubuntu# eksctl create nodegroup --cluster=aditya-eks \
--region=us-east-1 \
--name=node2 \
--node-type=t3.medium \
--nodes=3 \
--nodes-min=2 \
--nodes-max=4 \
--node-volume-size=20 \
--ssh-access \
--ssh-public-key=bms-key \
--managed \
--asg-access \
--external-dns-access \
--full-ecr-access \
--appmesh-access \
--alb-ingress-access
2026-04-22 09:00:59 [i] will use version 1.30 for new nodegroup(s) based on control plane version
2026-04-22 09:01:00 [i] nodegroup "node2" will use "" [AmazonLinux2023/1.30]
2026-04-22 09:01:00 [i] using EC2 key pair "bms-key"
2026-04-22 09:01:00 [i] 1 nodegroup (node2) was included (based on the include/exclude rules)
2026-04-22 09:01:00 [i] will create a CloudFormation stack for each of 1 managed nodegroups in cluster "aditya-eks"
2026-04-22 09:01:00 [i]
2 sequential tasks: { fix cluster compatibility, 1 task: { 1 task: { create managed nodegroup "node2" } } }
2026-04-22 09:01:00 [i] checking cluster stack for missing resources
2026-04-22 09:01:00 [i] cluster stack has all required resources
2026-04-22 09:01:00 [i] building managed nodegroup stack "eksctl-aditya-eks-nodegroup-node2"
```

STEP 22: CREATE A PIPELINE JOB AND CONFIGURE JENKINS PIPELINE

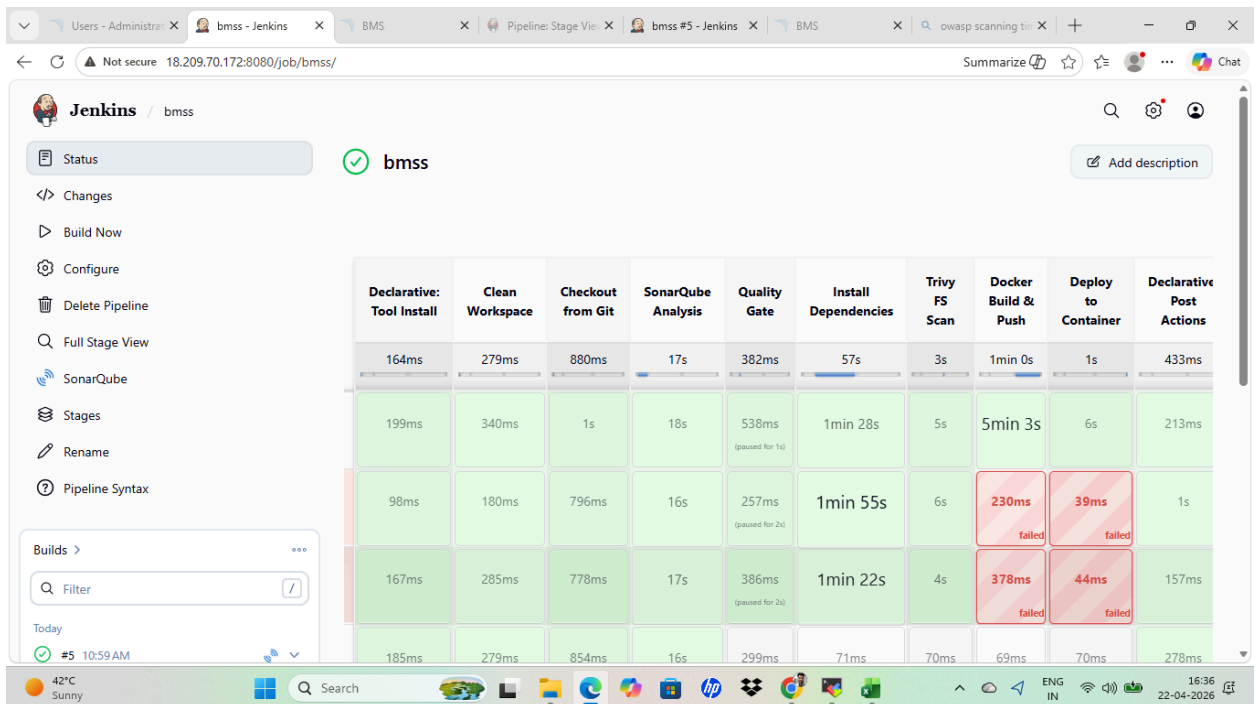


The screenshot shows the Jenkins 'Configure' page for a pipeline job named 'bms-app'. The left sidebar contains links for General, Triggers, Pipeline, and Advanced. The 'Pipeline' section is selected. The 'Definition' dropdown is set to 'Pipeline script'. A text area contains a Groovy script defining a pipeline with stages for tool installation, environment setup, and a 'Clean Workspace' stage. The script is as follows:

```
1 pipeline {
2   agent any
3   tools {
4     jdk 'jdk17'
5     nodejs 'node23'
6   }
7   environment {
8     SCANNER_HOME = tool 'sonar-scanner'
9   }
10  stages {
11    stage('Clean Workspace') {
12      steps {
13        cleanWs()
14      }
15    }
16  }
17 }
```

Buttons for 'Save' and 'Apply' are visible at the bottom of the script editor.

STEP 23: BUILD THE PIPELINE



The screenshot shows the Jenkins 'Builds' page for the 'bmss' pipeline. The page displays a table of build results for various stages. The 'bmss' job is marked as successful with a green checkmark. The table shows the following data:

Builds >	Declarative: Tool Install	Clean Workspace	Checkout from Git	SonarQube Analysis	Quality Gate	Install Dependencies	Trivy FS Scan	Docker Build & Push	Deploy to Container	Declarative Post Actions
10:59 AM	164ms	279ms	880ms	17s	382ms	57s	3s	1min 0s	1s	433ms
	199ms	340ms	1s	18s	538ms	1min 28s	5s	5min 3s	6s	213ms
	98ms	180ms	796ms	16s	257ms	1min 55s	6s	230ms failed	39ms failed	1s
	167ms	285ms	778ms	17s	386ms	1min 22s	4s	378ms failed	44ms failed	157ms
	185ms	279ms	854ms	16s	299ms	71ms	70ms	69ms	70ms	278ms

The table indicates that the 'Docker Build & Push' and 'Deploy to Container' stages failed in the second and third builds shown.

This is pipeline:

```
pipeline {
  agent any
  tools {
    jdk 'jdk17'
    nodejs 'node23'
  }
  environment {
    SCANNER_HOME = tool 'sonar-scanner'
  }
  stages {
    stage('Clean Workspace') {
      steps {
        cleanWs()
      }
    }
    stage('Checkout from Git') {
      steps {
        git branch: 'main', url:
'https://github.com/adityabob/OnlineTicketPlatform-BMS.git'
        sh 'ls -la' // Verify files after checkout
      }
    }
    stage('SonarQube Analysis') {
      steps {
        withSonarQubeEnv('sonar-server') {
          sh '''
            $SCANNER_HOME/bin/sonar-scanner -Dsonar.projectName=BMS \
            -Dsonar.projectKey=BMS
          '''
        }
      }
    }
    stage('Quality Gate') {
      steps {
        script {
          waitForQualityGate abortPipeline: false, credentialsId: 'sonar-token'
        }
      }
    }
    stage('Install Dependencies') {
      steps {
        sh '''
          cd bookmyshow-app
        '''
      }
    }
  }
}
```



```

ls -la # Verify package.json exists
if [ -f package.json ]; then
    rm -rf node_modules package-lock.json # Remove old dependencies
    npm install # Install fresh dependencies
else
    echo "Error: package.json not found in bookmyshow-app!"
    exit 1
fi
'''
}
}

stage('Trivy FS Scan') {
    steps {
        sh 'trivy fs . > trivyfs.txt'
    }
}

stage('Docker Build & Push') {
    steps {
        script {
            withDockerRegistry(credentialsId: 'docker-creds') {
                sh '''
                echo "Building Docker image..."
                docker build --no-cache -t adityabob135/bms:latest -f
bookmyshow-app/Dockerfile bookmyshow-app

                echo "Pushing Docker image to registry..."
                docker push adityabob135/bms:latest
                '''
            }
        }
    }
}

stage('Deploy to Container') {
    steps {
        sh '''
        echo "Stopping and removing old container..."
        docker stop bms || true
        docker rm bms || true

        echo "Running new container on port 3000..."
        docker run -d --restart=always --name bms -p 3000:3000
adityabob135/bms:latest
        '''
    }
}

```

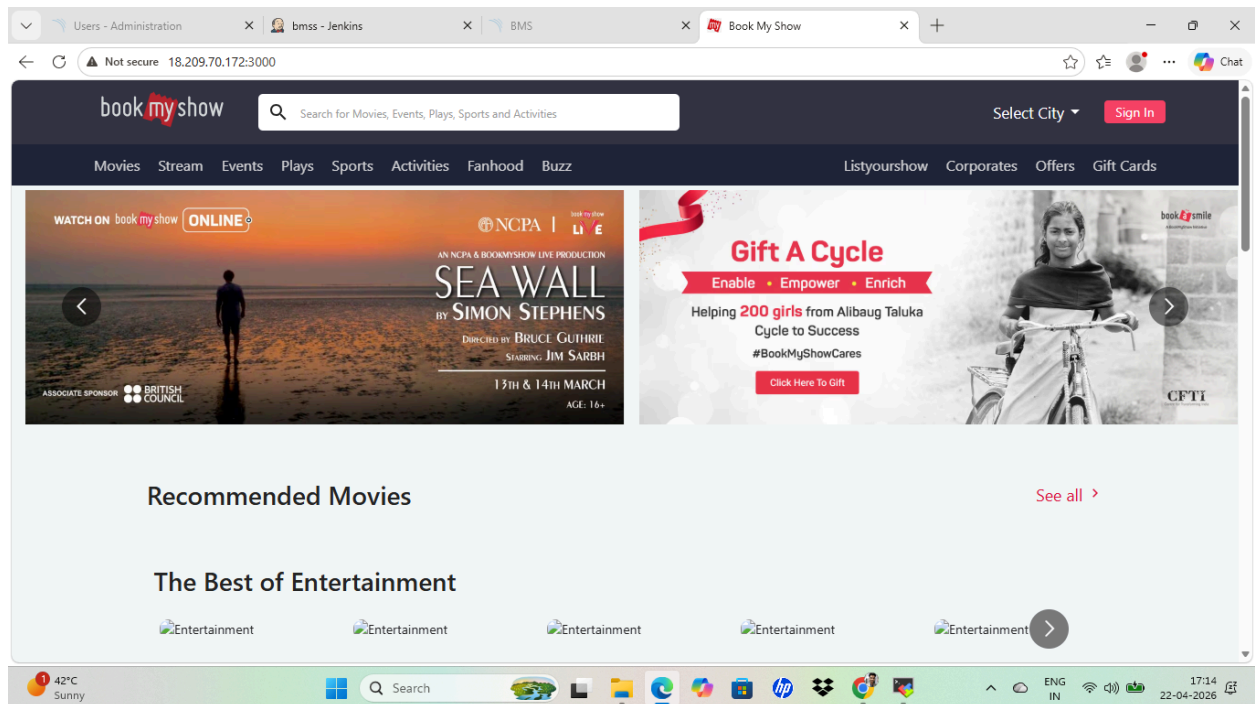
```

    echo "Checking running containers..."
    docker ps -a

    echo "Fetching logs..."
    sleep 5 # Give time for the app to start
    docker logs bms
    ""
  }
}
}
post {
  always {
    emailx attachLog: true,
    subject: "${currentBuild.result}",
    body: "Project: ${env.JOB_NAME}<br/>" +
      "Build Number: ${env.BUILD_NUMBER}<br/>" +
      "URL: ${env.BUILD_URL}<br/>",
    to: 'adityabobade135@gmail.com',
    attachmentsPattern: 'trivyfs.txt,trivyimage.txt'
  }
}
}
}

```

Step 24: we can access the website using browser



Step 25: all the cluster nodes are created and we can see it in the aws

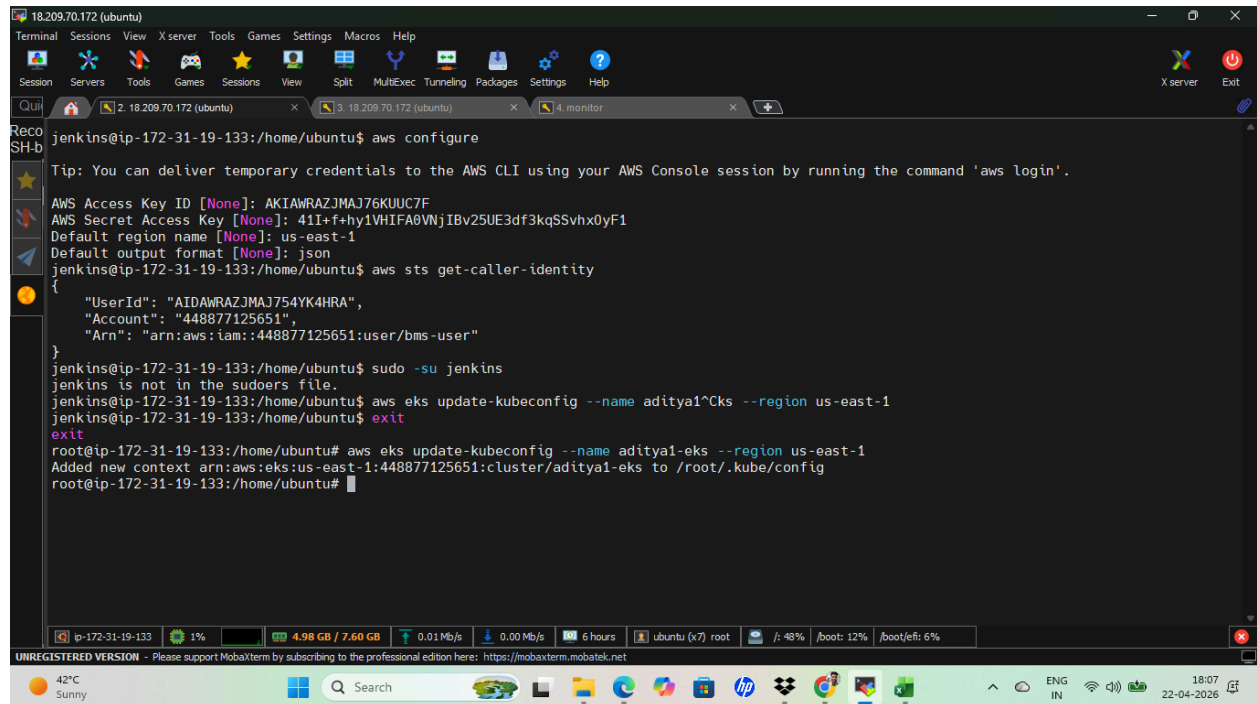
The screenshot displays the AWS Management Console for the 'us-east-1' region. The 'Instances' page shows five EC2 instances, all in a 'Running' state. The instances are:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
aditya1-eks-n...	i-070ae0970090c912e	Running	t3.small	3/3 checks passed	View alarms +	us-east-1a
aditya1-eks-n...	i-08142f92e367eb81b	Running	t3.small	3/3 checks passed	View alarms +	us-east-1a
book-my-show	i-02f792b0c1d6f35d4	Running	m7i-flex.large	3/3 checks passed	View alarms +	us-east-1c
monitoring ins...	i-0415203be031963f8	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b
aditya1-eks-n...	i-00b41fa37e446abe3	Running	t3.small	3/3 checks passed	View alarms +	us-east-1b

Below the instances list, a terminal window is open, showing the output of the 'ps aux | grep jenkins' command. The output indicates that Jenkins is running on the 'aditya1-eks' cluster nodes. The terminal also shows the 'sudo systemctl status jenkins' command being executed on the 'aditya1-eks' cluster nodes, confirming that Jenkins is running successfully.

```
root@ip-172-31-19-133:/home/ubuntu# ps aux | grep jenkins
root 7551 0.0 0.0 17148 5184 pts/0 T 06:35 0:00 sudo systemctl status jenkins
root 7552 0.0 0.0 17148 2420 pts/1 Ss 06:35 0:00 sudo systemctl status jenkins
root 7553 0.0 0.0 17640 4556 pts/1 T+ 06:35 0:00 systemctl status jenkins
root 99589 0.0 0.0 17560 5196 pts/3 T 10:15 0:00 sudo systemctl restart jenkins
root 99590 0.0 0.0 17560 2540 pts/4 Ss 10:15 0:00 sudo systemctl restart jenkins
root 99591 0.0 0.0 17640 4248 pts/4 T+ 10:15 0:00 systemctl restart jenkins
jenkins 100023 1.8 12.3 4780692 982064 ? Ssl 10:16 2:27 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --web
root=/var/cache/jenkins/war --httpPort=8080
root 153503 0.0 0.0 7084 2292 pts/3 S+ 12:30 0:00 grep --color=auto jenkins
root@ip-172-31-19-133:/home/ubuntu# sudo -su jenkins
jenkins@ip-172-31-19-133:/home/ubuntu$ whoami
jenkins
jenkins@ip-172-31-19-133:/home/ubuntu$
```

Step 26: configure aws in the node cluster

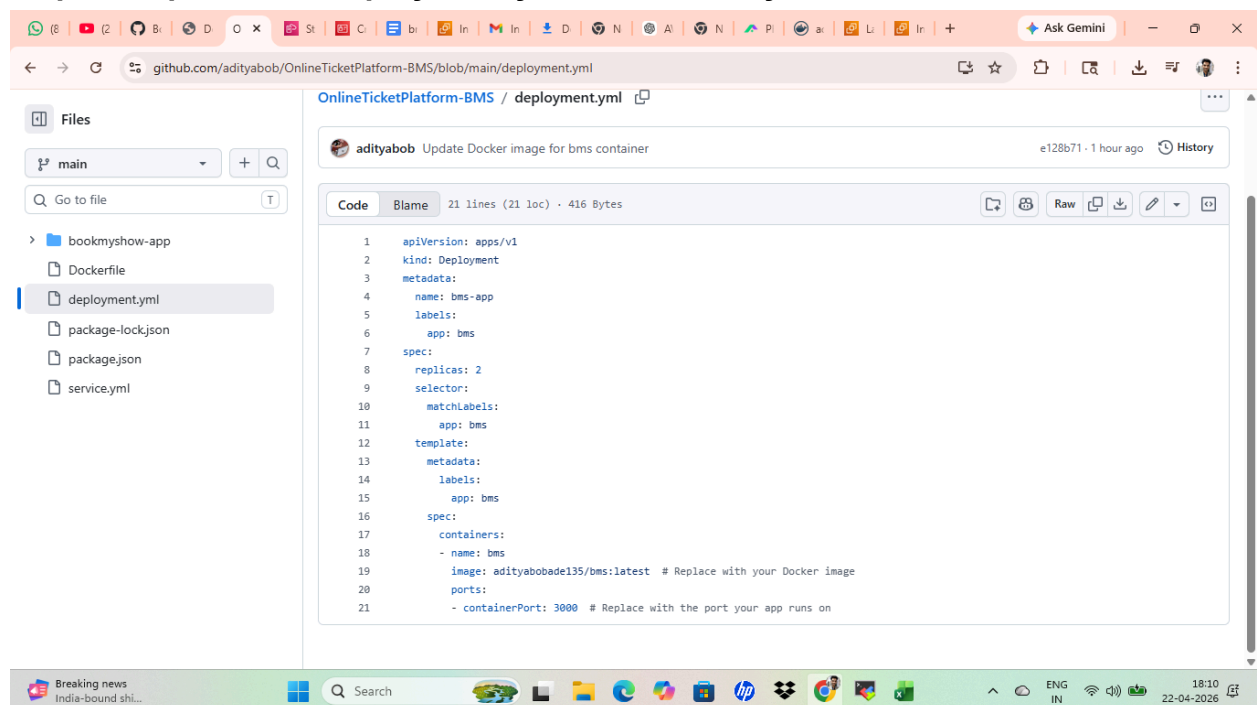


The screenshot shows a terminal window with the following commands and output:

```
jenkins@ip-172-31-19-133:/home/ubuntu$ aws configure
Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

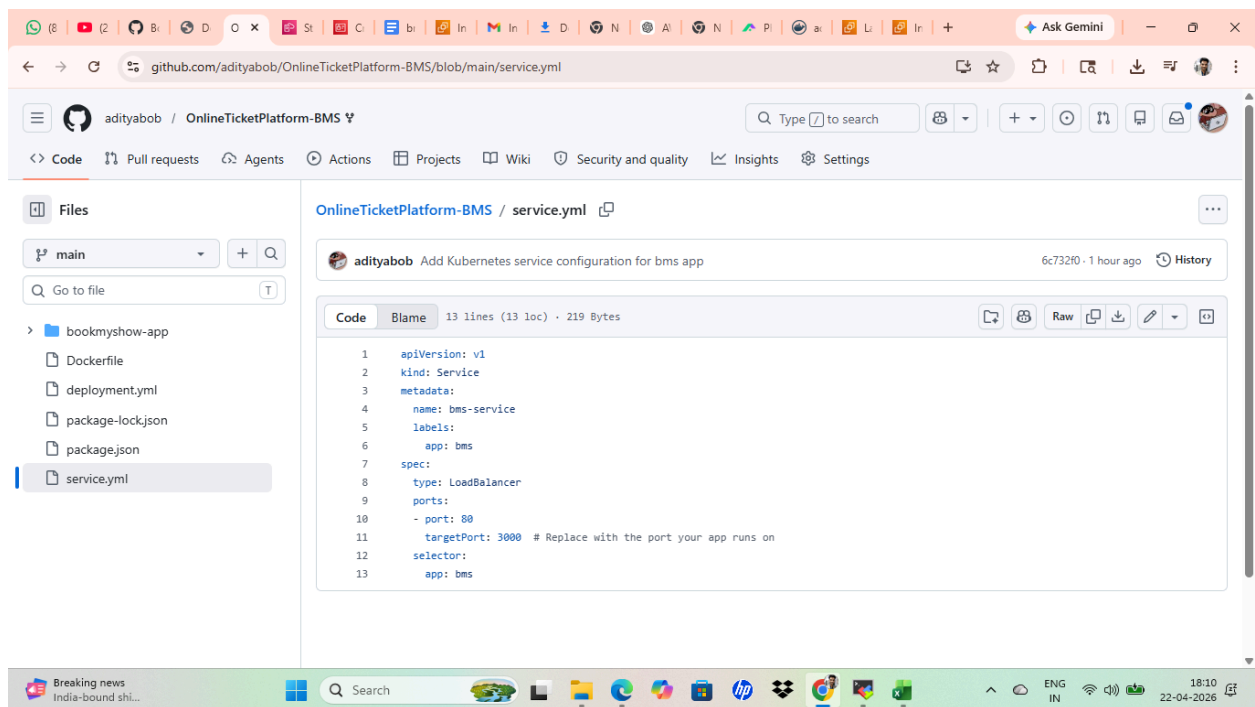
AWS Access Key ID [None]: AKIAWRAZJMAJ76KUUC7F
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
jenkins@ip-172-31-19-133:/home/ubuntu$ aws sts get-caller-identity
{
  "UserId": "AIDAWRAZJMAJ754YK4HRA",
  "Account": "448877125651",
  "Arn": "arn:aws:iam::448877125651:user/bms-user"
}
jenkins@ip-172-31-19-133:/home/ubuntu$ sudo -su jenkins
jenkins is not in the sudoers file.
jenkins@ip-172-31-19-133:/home/ubuntu$ aws eks update-kubeconfig --name aditya1^Cks --region us-east-1
jenkins@ip-172-31-19-133:/home/ubuntu$ exit
root@ip-172-31-19-133:/home/ubuntu# aws eks update-kubeconfig --name aditya1-eks --region us-east-1
Added new context arn:aws:eks:us-east-1:448877125651:cluster/aditya1-eks to /root/.kube/config
root@ip-172-31-19-133:/home/ubuntu#
```

Step 27 : update the deployment.yml and service.yml file

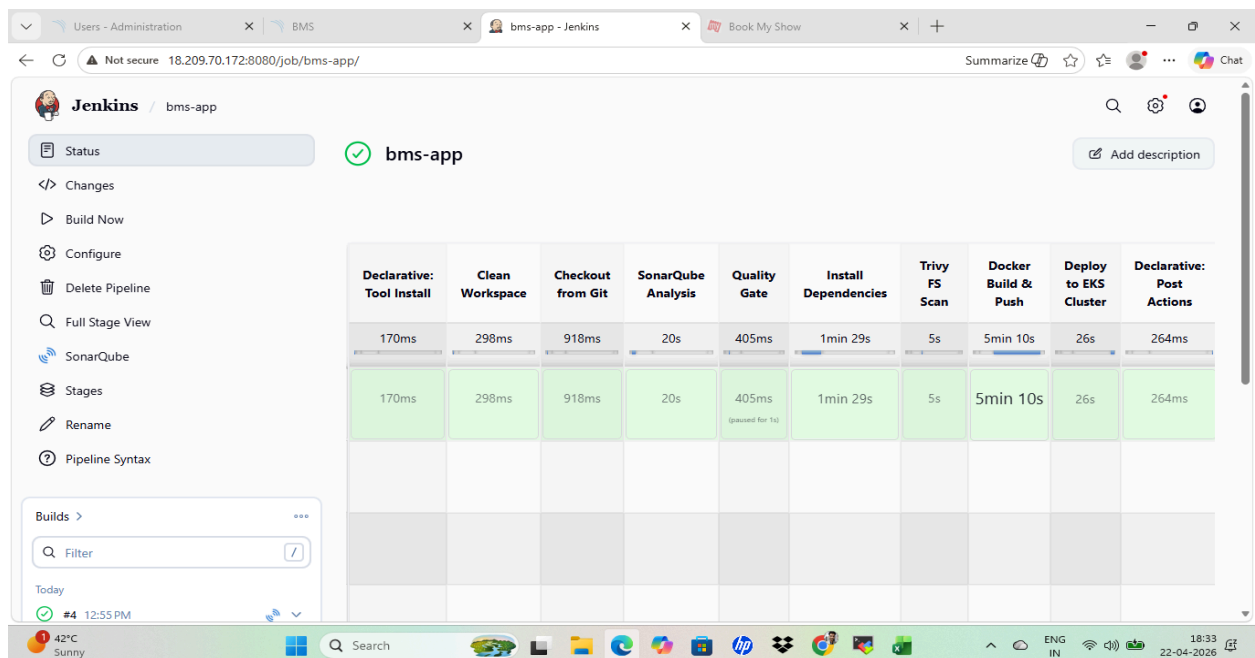


The screenshot shows a GitHub repository page for 'OnlineTicketPlatform-BMS'. The file 'deployment.yml' is selected, showing the following content:

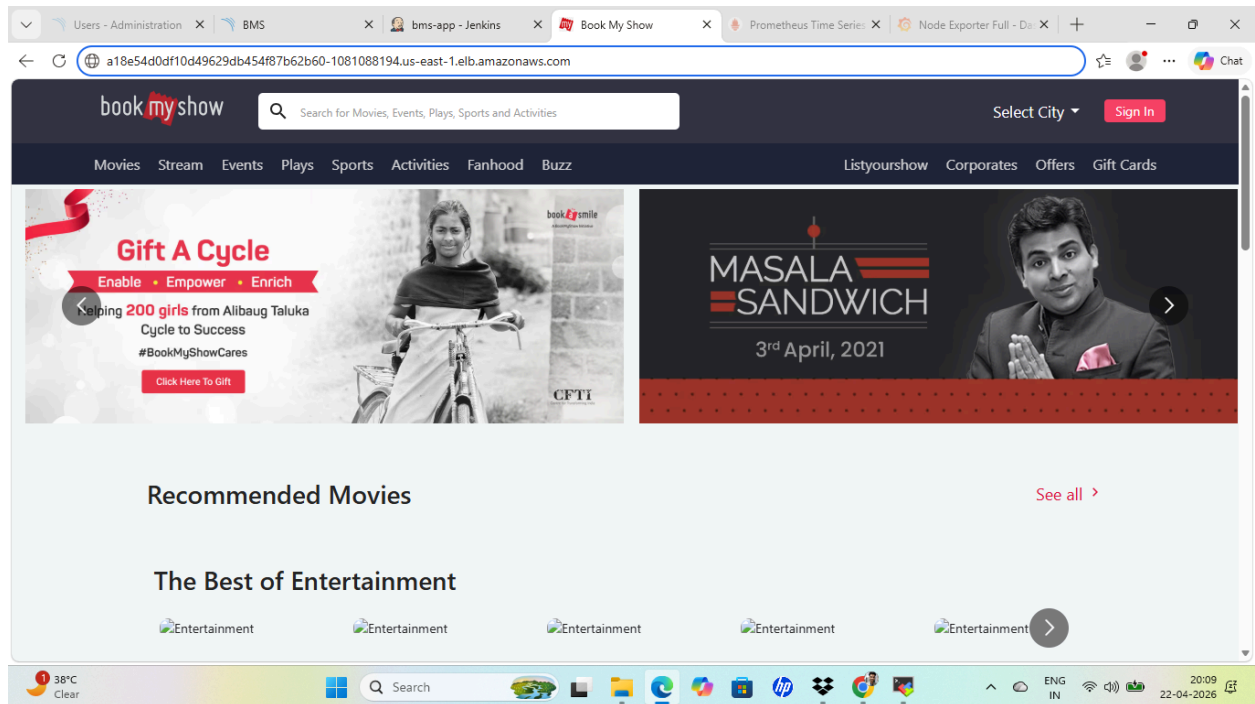
```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: bms-app
5    labels:
6      app: bms
7  spec:
8    replicas: 2
9    selector:
10     matchLabels:
11       app: bms
12    template:
13     metadata:
14       labels:
15         app: bms
16     spec:
17       containers:
18         - name: bms
19           image: adityabobade135/bms:latest # Replace with your Docker image
20           ports:
21             - containerPort: 3000 # Replace with the port your app runs on
```



STEP 28: configure the pipeline for kubernetes cluster and build it

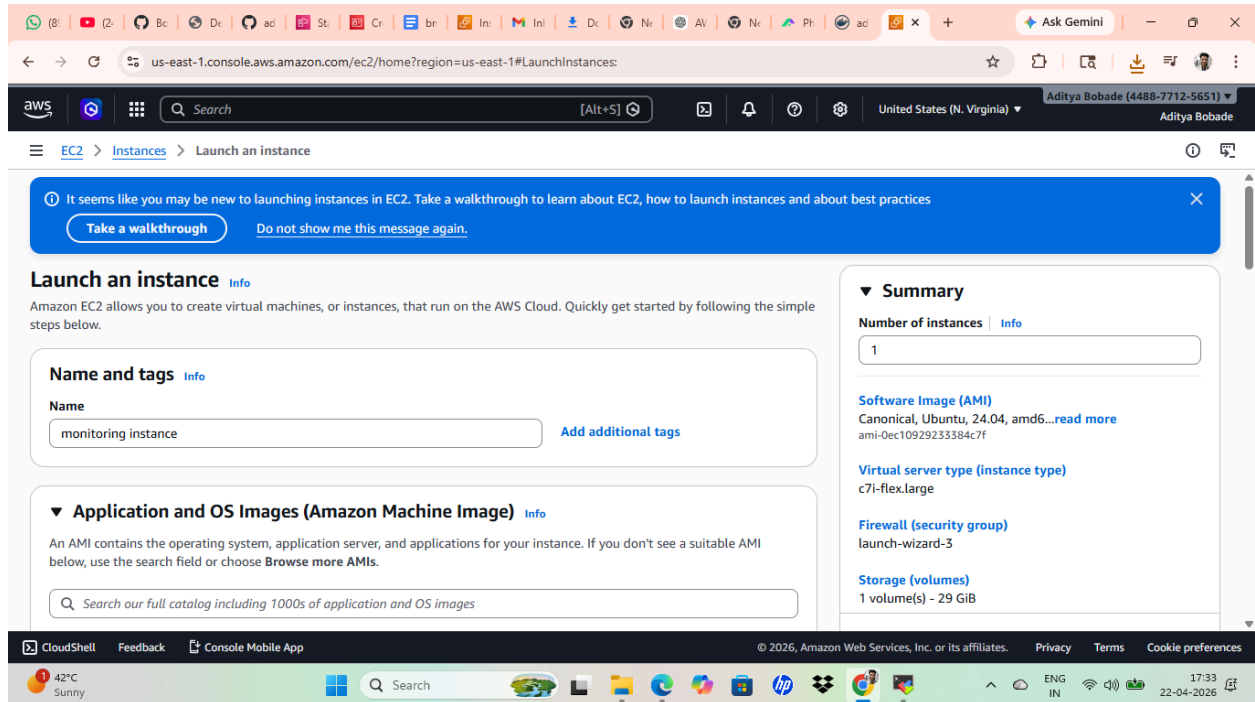


Step 29: you can access the website ussing the load balancer url

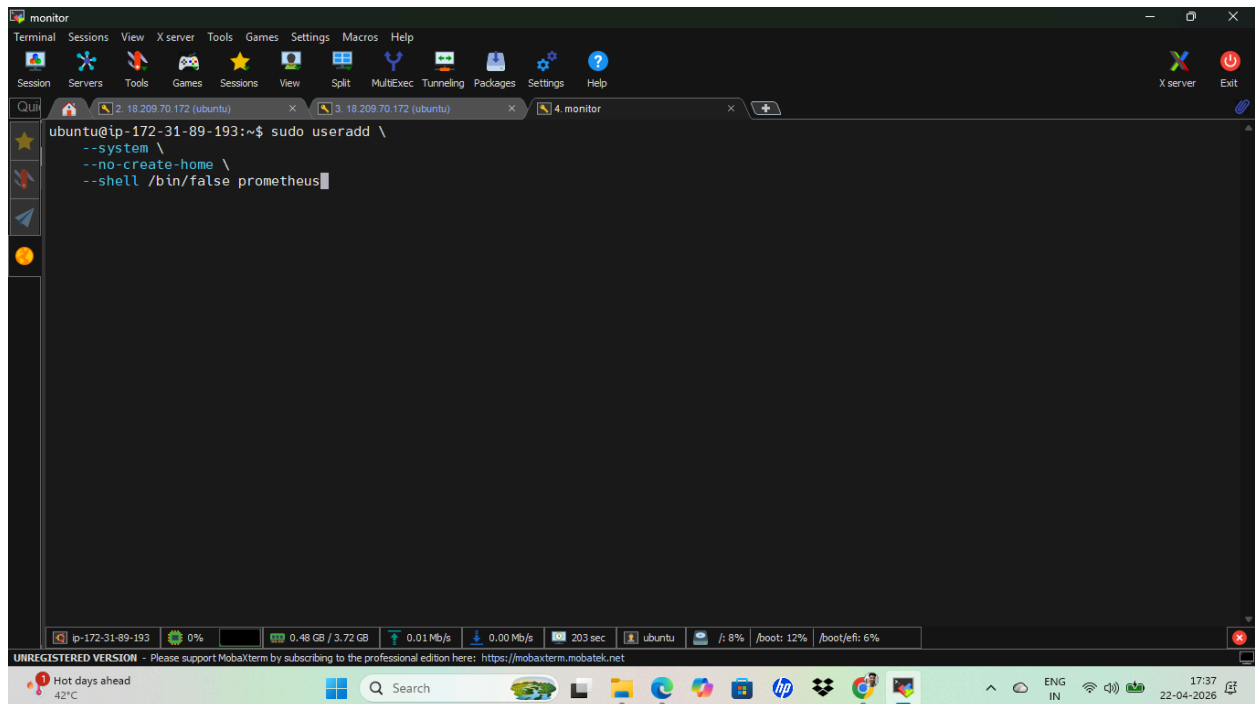


MONITORING PART:

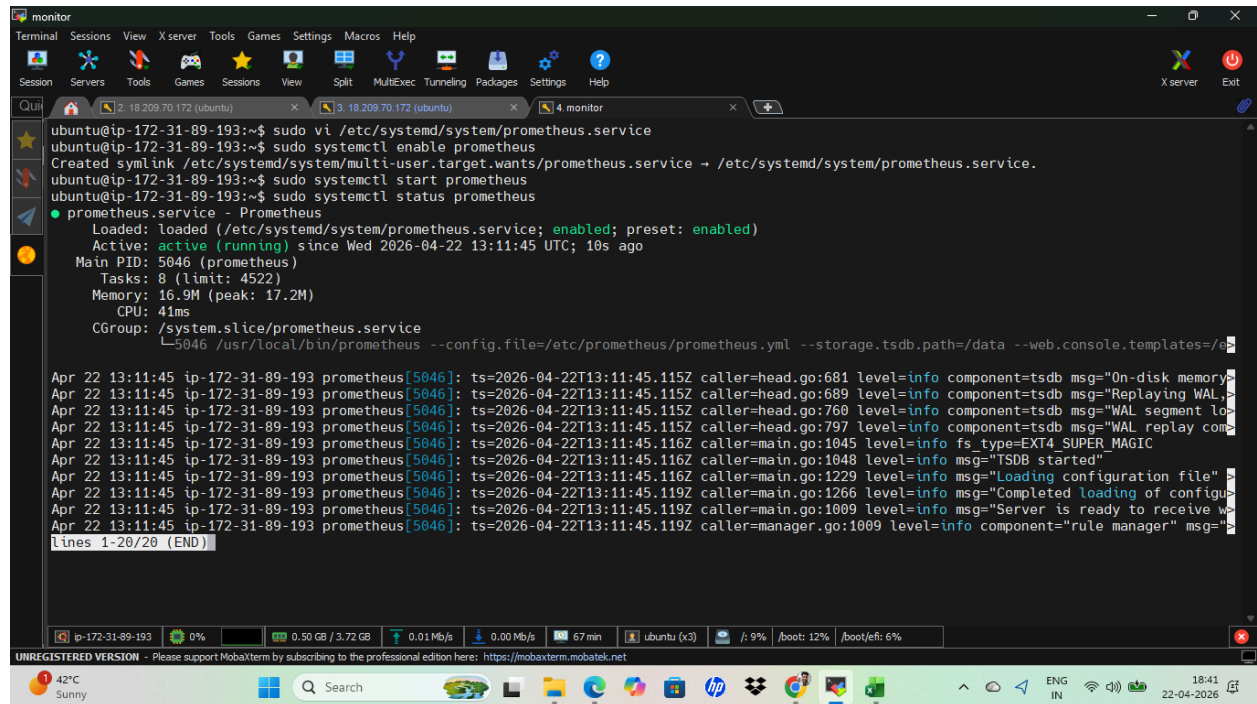
Step 1: create an instance dedicated to monitoring where we would be setup prometheus & grafana



Step 2: Add prometheus user to the instance



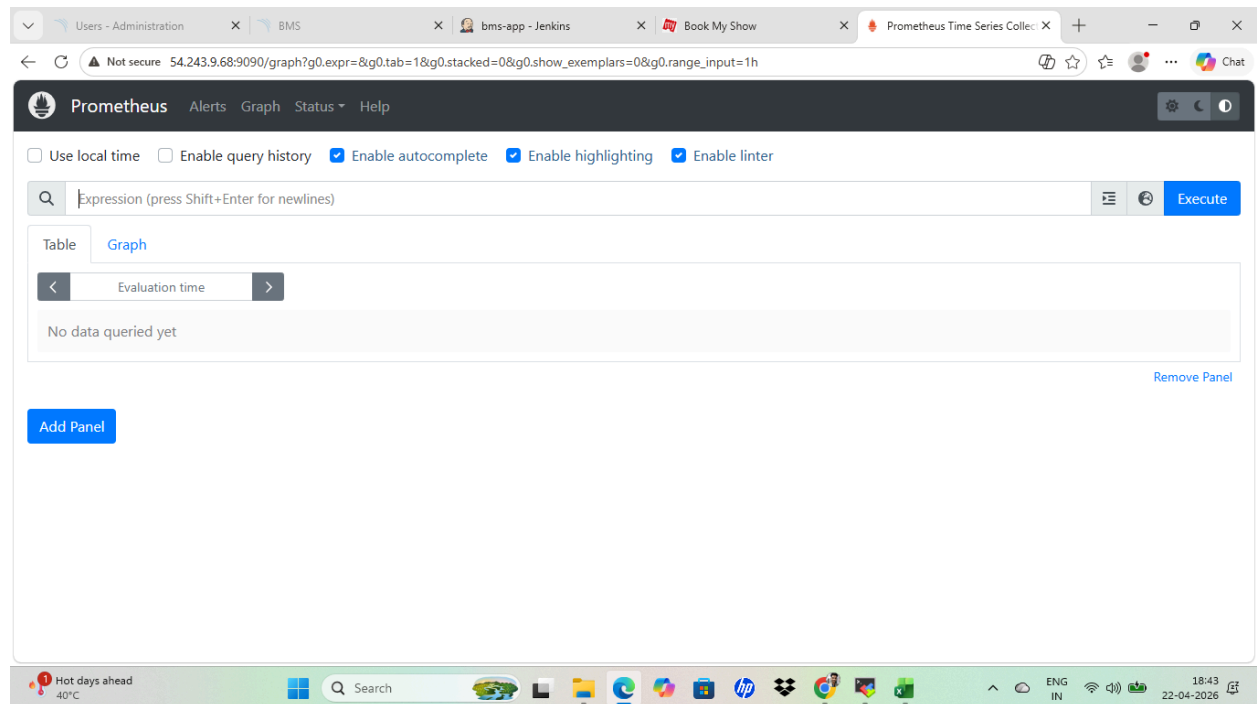
Step 3: do all the required setup and install prometheus



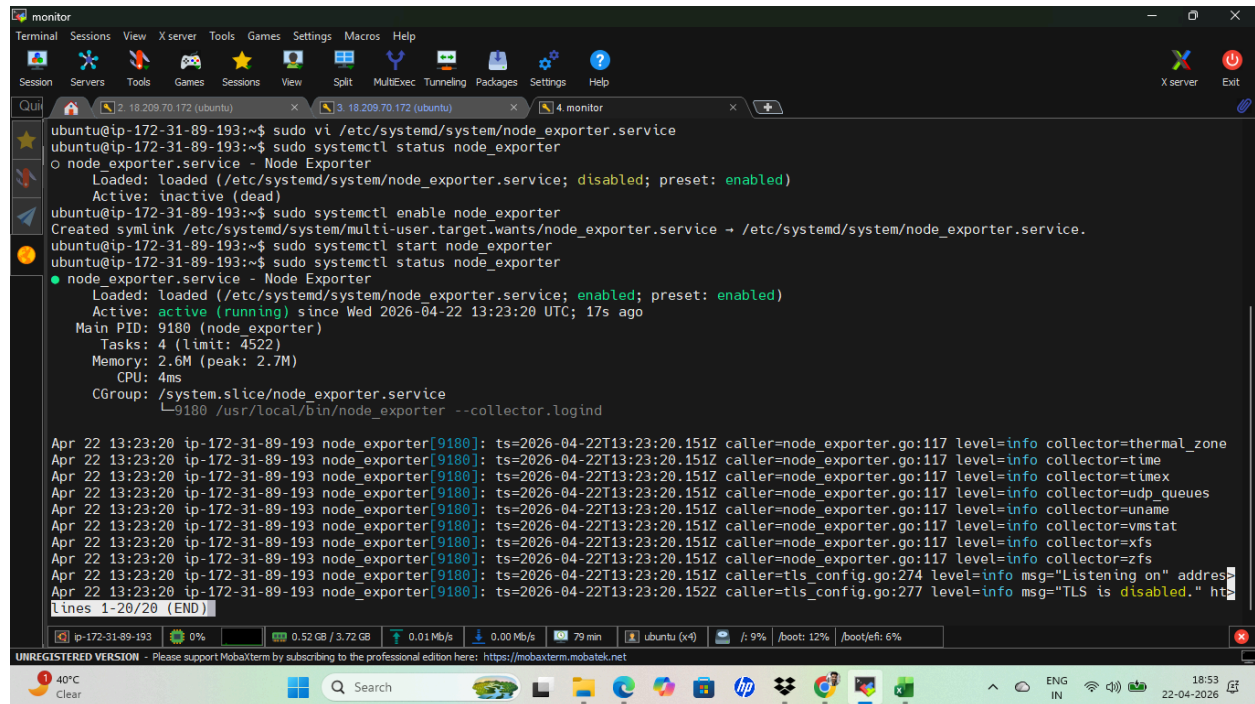
```
ubuntu@ip-172-31-89-193:~$ sudo vi /etc/systemd/system/prometheus.service
ubuntu@ip-172-31-89-193:~$ sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
ubuntu@ip-172-31-89-193:~$ sudo systemctl start prometheus
ubuntu@ip-172-31-89-193:~$ sudo systemctl status prometheus
● prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 13:11:45 UTC; 10s ago
     Main PID: 5046 (prometheus)
        Tasks: 8 (limit: 4522)
       Memory: 16.9M (peak: 17.2M)
          CPU: 41ms
       CGroup: /system.slice/prometheus.service
               └─5046 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/data --web.console.templates=/e

Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.115Z caller=head.go:681 level=info component=tsdb msg="On-disk memory-
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.115Z caller=head.go:689 level=info component=tsdb msg="Replaying WAL-
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.115Z caller=head.go:760 level=info component=tsdb msg="WAL segment lo
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.115Z caller=head.go:797 level=info component=tsdb msg="WAL replay com
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.116Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.116Z caller=main.go:1048 level=info msg="TSDB started"
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.116Z caller=main.go:1229 level=info msg="Loading configuration file"
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.119Z caller=main.go:1266 level=info msg="Completed loading of configu
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.119Z caller=main.go:1009 level=info msg="Server is ready to receive w
Apr 22 13:11:45 ip-172-31-89-193 prometheus[5046]: ts=2026-04-22T13:11:45.119Z caller=manager.go:1009 level=info component="rule manager" msg=">
```

Step 4: Access the prometheus in browser public ip of instance and port no of prom,



Step 5: Install node exporter in the monitoring instance



```
monitor
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quit 18.209.70.172 (ubuntu) 18.209.70.172 (ubuntu) 4 monitor
ubuntu@ip-172-31-89-193:~$ sudo vi /etc/systemd/system/node_exporter.service
ubuntu@ip-172-31-89-193:~$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; disabled; preset: enabled)
   Active: inactive (dead)
ubuntu@ip-172-31-89-193:~$ sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.
ubuntu@ip-172-31-89-193:~$ sudo systemctl start node_exporter
ubuntu@ip-172-31-89-193:~$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 13:23:20 UTC; 17s ago
     Main PID: 9180 (node_exporter)
       Tasks: 4 (limit: 4522)
      Memory: 2.6M (peak: 2.7M)
         CPU: 4ms
        CGroup: /system.slice/node_exporter.service
                └─9180 /usr/local/bin/node_exporter --collector.logind

Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=thermal_zone
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=time
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=timex
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=udp_queues
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=uname
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=vmstat
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=xfs
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=node_exporter.go:117 level=info collector=zfs
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.151Z caller=tls_config.go:274 level=info msg="Listening on" address=
Apr 22 13:23:20 ip-172-31-89-193 node_exporter[9180]: ts=2026-04-22T13:23:20.152Z caller=tls_config.go:277 level=info msg="TLS is disabled." http2
lines 1-26/20 (END)
ip-172-31-89-193 0% 0.52 GB / 3.72 GB 0.01 Mb/s 0.00 Mb/s 79 min ubuntu (x9) / 9% /boot: 12% /boot/efi: 6%
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

Step 6: Configure the the ip address of jenkins and prometheus in prometheus.yml

- job_name: 'node_exporter'
 - static_configs:
 - targets: ['54.243.9.68:9100']
- job_name: 'jenkins'
 - metrics_path: '/prometheus'
 - static_configs:
 - targets: ['18.209.70.172:8080']

```
monitor
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
ubuntu@ip-172-31-89-193:~$ cd /etc/prometheus/
ubuntu@ip-172-31-89-193:/etc/prometheus$ ls
console libraries consoles prometheus.yml
ubuntu@ip-172-31-89-193:/etc/prometheus$ sudo vi prometheus.yml
ubuntu@ip-172-31-89-193:/etc/prometheus$ promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax
ubuntu@ip-172-31-89-193:/etc/prometheus$
```

Step 7: you can see prometheus targets in prometheus server

Users - Administration | BMS | bms-app - Jenkins | Book My Show | Prometheus Time Series Collec: X

Not secure 54.243.9.68:9090/targets?search=#pool-jenkins

Prometheus Alerts Graph Status Help

Targets

All scrape pools | All | Unhealthy | Collapse All | Filter by endpoint or labels | Unknown | Unhealthy | Healthy

jenkins (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://18.209.70.172:8080/prometheus	UP	instance="18.209.70.172:8080" job="jenkins"	18.72s ago	36.304ms	

node_exporter (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://54.243.9.68:9100/metrics	UP	instance="54.243.9.68:9100" job="node_exporter"	15.232s ago	16.538ms	

prometheus (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	22.905s ago	3.056ms	

40°C Clear | Search | ENG IN | 19:05 22-04-2026

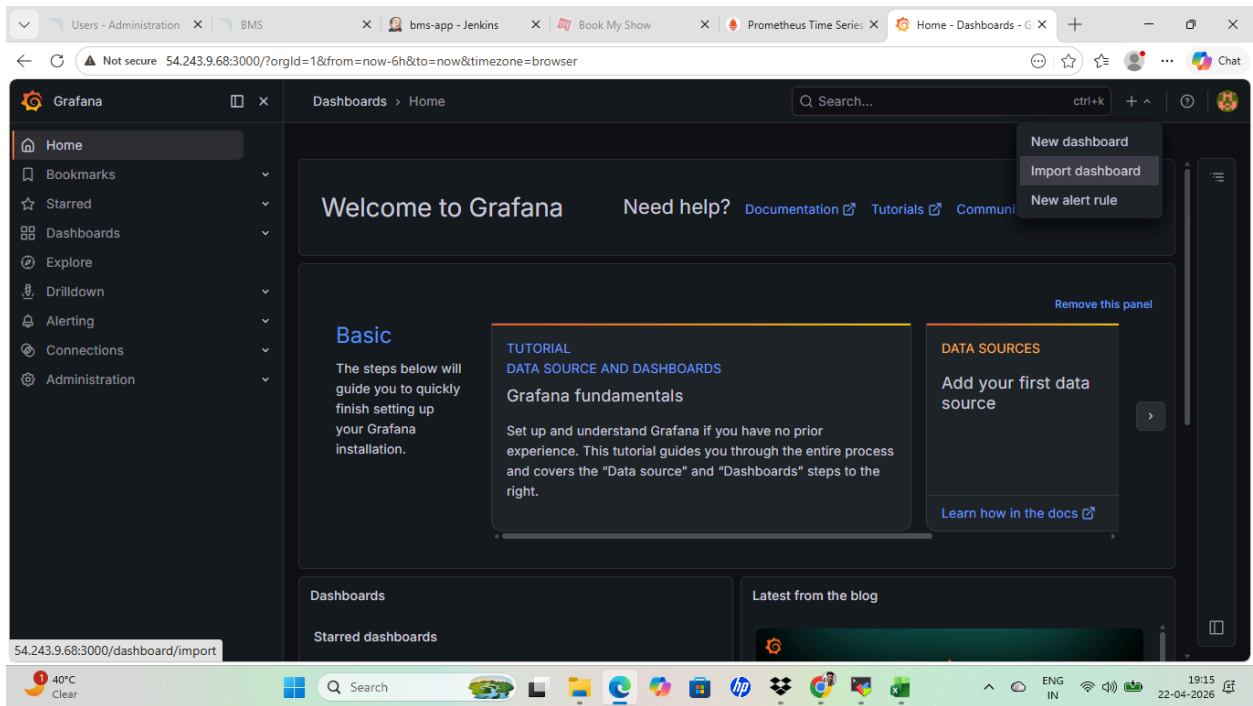
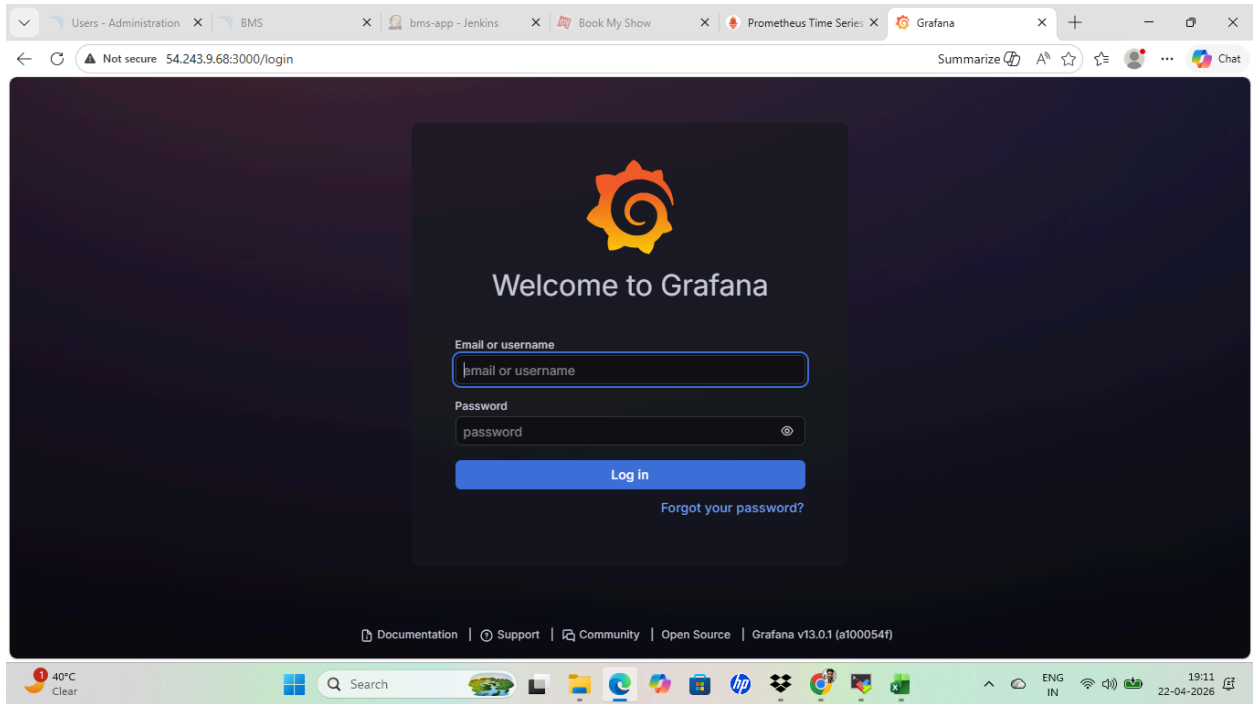
Step 8: Install grafana in monitoring instance

```
monitor
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick 1 18.209.70.172 (ubuntu) x 3 18.209.70.172 (ubuntu) x 4 monitor x +
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
ubuntu@ip-172-31-89-193:~$ echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a /etc/apt/sources.list.d/grafana.list
deb https://packages.grafana.com/oss/deb stable main
ubuntu@ip-172-31-89-193:~$ Update the package list and install Grafana:
sudo apt-get update
sudo apt-get -y install grafana
Update: command not found
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://packages.grafana.com/oss/deb stable InRelease [7661 B]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://packages.grafana.com/oss/deb stable/main amd64 Packages [489 kB]
Fetched 497 kB in 0s (1086 kB/s)
Reading package lists... Done
W: https://packages.grafana.com/oss/deb/dists/stable/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPR
ECATION section in apt-key(8) for details.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  grafana
0 upgraded, 1 newly installed, 0 to remove and 60 not upgraded.
Need to get 274 MB of archives.
After this operation, 1028 MB of additional disk space will be used.
Get:1 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 13.0.1 [274 MB]
Fetched 274 MB in 5s (55.5 MB/s)
Selecting previously unselected package grafana.
(Reading database ... 72570 files and directories currently installed.)
```

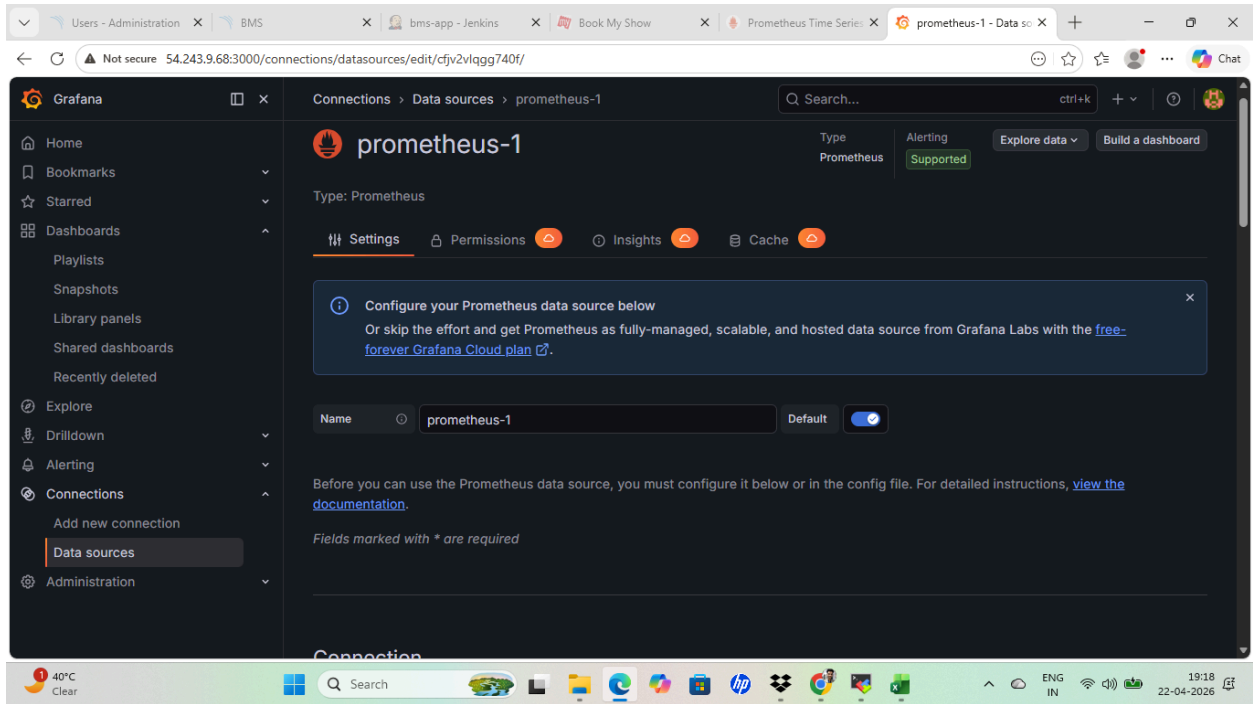
```
monitor
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick 1 18.209.70.172 (ubuntu) x 3 18.209.70.172 (ubuntu) x 4 monitor x +
ubuntu@ip-172-31-89-193:~$ sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
ubuntu@ip-172-31-89-193:~$ sudo systemctl start grafana-server
ubuntu@ip-172-31-89-193:~$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-22 13:40:31 UTC; 9s ago
     Docs: http://docs.grafana.org
   Main PID: 16654 (grafana)
    Tasks: 13 (limit: 4522)
   Memory: 75.7M (peak: 76.0M)
      CPU: 1.451s
   CGroup: /system.slice/grafana-server.service
           └─16654 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packag
             16703 /usr/share/grafana/data/plugins-bundled/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64

Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert.notifier t=2026-04-22T13:40:39.571183942Z level=info component=alertmanager org=
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert.notifier component=alertmanager orgID=1 t=2026-04-22T13:40:39.572799797Z level=
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert.notifier t=2026-04-22T13:40:39.572832557Z level=info component=alertmanager org=
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert.writer t=2026-04-22T13:40:39.587275057Z level=info msg="Setting up remote write
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert t=2026-04-22T13:40:39.587470203Z level=info msg="Using protobuf-based alert ins
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=ngalert.state.manager.persist t=2026-04-22T13:40:39.587489603Z level=info msg="Using sys
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=query_data t=2026-04-22T13:40:39.588116128Z level=info msg="Query Service initializatio
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=annotation.app t=2026-04-22T13:40:39.614092704Z level=info msg="Starting annotation cle
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=infra.usagstats.collector t=2026-04-22T13:40:39.616100895Z level=info msg="registering
Apr 22 13:40:39 ip-172-31-89-193 grafana[16654]: logger=server t=2026-04-22T13:40:39.616224578Z level=info msg="Writing PID file" path=/run/graf
lines 1-22/22 (END)
```

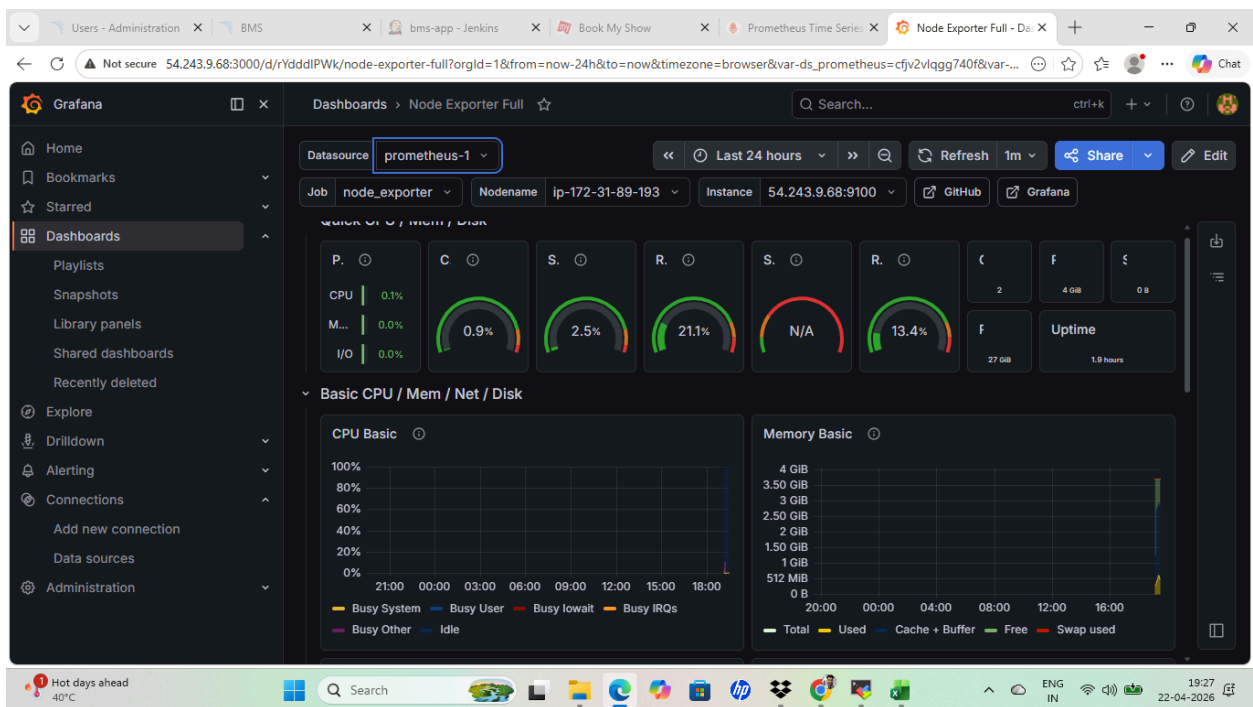
Step 9: access the grafana server using public ip and port no



Step 10: configure prometheus url with grafana in data sources



Step 11: import dashboards where we can see cpu ,ram and memory usage



Users - Administration × BMS × bms-app - Jenkins × Book My Show × Prometheus Time Series × Node Exporter Full - D × +

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